Adaptive stability of the Black alder ecoepidemic system: integrated intervention alone prevents regime shift under phenological variability

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Abstract

Black alder (Alnus glutinosa), the phyllophagous leaf beetle (Agelastica alni), the koinobiont larval endoparasitoid (Meigenia mutabilis), and generalist passeriform birds constitute an ecoepidemic system across alluvial forests in Europe. Phyllophagy by beetle larvae causes foliar biomass loss in the tree. The endoparasitoid parasitises susceptible beetle larvae by ovipositing on their integument; emergence of adult parasitoids kills parasitised larvae, providing natural regulation of defoliation. Generalist passeriform birds, following optimal foraging strategies, also contribute to biocontrol by preferentially consuming the most abundant prey during outbreaks. Induced phytochemical changes in pest-infected trees provide stand-level resistance that may accumulate and persist into the subsequent year. Thus, resistance differs across intra-annual and interannual timescales, while vernal regrowth, providing system resilience via canopy recovery, is observable only at an annual timescale. This temporal discordance introduces complexity in system stability. This study uses a process-based hybrid seasonal–annual model of this focal ecoepidemic system to quantify how resistance and resilience jointly structure stability across nine phenological scenarios associated with climatic variation, and applies optimal control theory to contrast three pest-management strategies. Results show that the coexistence equilibrium is the least frequently stable regime across phenological parameter space, while the parasitoid-free state is the most frequent. When all phenological parameters vary simultaneously, the probability of regime shift from coexistence reaches 81.5%. Among the three management strategies tested, only the fully integrated strategy combining parasitoid augmentation, targeted larval removal, and bird-habitat enhancement achieves a locally stable managed equilibrium and does so at minimum cost. These findings indicate that maintaining parasitoid-based biological control requires active, multi-component intervention and that single-agent strategies, though capable of short-term suppression, fail to ensure long-term system stability.

Number of Tables and Figures: 7 figures and 1 table in the main manuscript.

1. Introduction

Black alder (*Alnus glutinosa*) dominates European riparian zones and supports many flora and fauna habitats (FFH; European Commission, 1992). Outbreaks of the phyllophagous leaf beetle *Agelastica alni* can substantially defoliate the canopy (Dolch & Tscharntke, 2000). If defoliation recurs, repeated outbreaks may cause forest system collapse due to its vulnerability to other environmental perturbations (Valor et al., 2020). When beetles attack, black alder shows induced physiological changes that may offer stand-level resistance (Dolch & Tscharntke, 2000). Two natural beetle enemies also help regulate their population. One is Meigenia mutabilis, a koinobiont larval endoparasitoid that parasitises beetle larvae (Dolch & Tscharntke, 2000; Stireman et al., 2006). Evidence of specialist birds controlling leaf beetle directly is rare in the literature. However, optimal foraging theory suggests generalist passeriform birds in these forests will feed on the beetles during outbreaks, targeting the most abundant resource (Remmel et al., 2011; Monteagudo et al., 2023). Although empirical evidence suggests reflushing of leaves after defoliation (Dolch & Tscharntke, 2000), repeated outbreaks can potentially constrain the recovery of the whole canopy at the forest level; therefore, it is expected to occur mostly through vernal growth during the next spring following the outbreak season. To prevent canopy loss from the beetle, often, Black alder-dominated habitats restoration involves management intervention, such as beetle removal and parasitoid introduction. The top-down control via optimal foraging by generalist passeriform birds and seasonal growth of each trophic level of this system provide a natural regulation that forms an ecological core of this system (Remmel et al., 2011; Monteagudo et al., 2023). The conversion from susceptible to parasitised beetle larvae forms the epidemic core of the system (Stireman et al., 2006). Overall, this ecoepidemic system offers an opportunity to investigate how natural regulation and stand-level resistance contribute to the system's stability, coupled with resilience from canopy recovery. Understanding the adaptability in this stability from dynamic resistance and resilience can offer insight into optimising the intervention strategies in a changing environment.

This leaf beetle typically causes outbreaks in spring and early summer. During this period, newly flushed alder foliage provides abundant food for larval feeding and development (Sivrikaya et al., 2020). In response, alder leaves undergo induced phytochemical and physical changes, including increased phenolic content, reduced starch, and the development of traits that deter beetles. These defences determine stand-level resistance and help prevent beetle spread to other susceptible trees (Tscharntke et al., 2001; Dolch & Tscharntke, 2000). At the same time, the koinobiont larval endoparasitoid tracks beetle larval aggregations and parasitises susceptible larvae by ovipositing on their integument. The subsequent emergence of adult parasitoids results in the death of parasitised larvae (Dolch & Tscharntke, 2000; Stireman et al., 2006). Notably, there is no well-studied defence from beetle larvae to the parasitoid. However, beetles possess chemical defences, such as foul odours, that deter specialist birds (Pasteels et al., 1995). While specialist birds rarely forage on this beetle, larvae remain vulnerable to generalist passeriform birds, which select the most abundant prey based on minimal search time (Stephens & Krebs, 1986). All these phenological scenarios relate to climatic variation and human interventions.

There are at least nine well-documented variable phenologies in this system. The length of the spring–summer season, which changes under a warming climate, determines the vernal growth period of the tree (Renner & Zohner, 2018). This also sets the vulnerability window when beetle larvae are exposed to parasitoids. Vernal growth of black alder controls canopy recovery from past defoliation and supports annual resilience (Dolch & Tscharntke, 2000). Early spring can cause faster recovery and lead to higher resilience. Winter severity influences the overwintering survival of both beetle and parasitoid (Sivrikaya et al., 2020; Stireman et al., 2006). This, in turn, determines the initial density of the susceptible beetle population. It also affects the initial density of parasitoids that can lay eggs and kill beetles, which together determine effective outbreak intensity for the following season. Parasitoid development rate varies with temperature and climate (Stireman et al., 2006). As a result, the mortality rate of parasitised beetle larvae can increase, while the mortality rate of susceptible larvae may decrease with a warmer climate. Aggregate predation pressure on the beetle from generalist passeriform birds also varies widely (Monteagudo et al., 2023). This depends on habitat structure, breeding disturbance, bird community composition, learned avoidance of beetle odour, and bird-habitat enhancement. Foliage growth changes due to accumulated phytochemical defence in frequently attacked trees (Tscharntke et al., 2001; Dolch

& Tscharntke, 2000). This can lead to lower carrying capacity for foraging aggregation of beetles. Decreasing foraging aggregation can lead to less parasitoid attraction toward beetles, and thus a lower parasitoid attack rate (Stireman et al., 2006). Therefore, differential variability in multiple phenologies within the same system can produce complex and adaptive system stability via dynamic resistance and resilience.

The adaptive stability from the variable phenologies can offer strategic planning in habitat restoration and especially pest control intervention. The optimal control theory can help minimise the cost of intervention strategies using the interdependence of these phenologies, intervention measures, and environmental changes (Lenhart & Workman, 2007). However, in order to find the optimal strategies of pest management intervention, subject to phenological scenarios, understanding possible stable equilibria and their desirability is also necessary. For instance, since there is no direct empirical evidence on the behavioural or visual distinction of the parasitised beetles from susceptible ones, the consumption of both classes can be assumed to be equal by generalist passeriform birds (Segoli et al., 2024). Under this assumption, bird predation on parasitised larvae can be interpreted as intraguild predation; foraging on parasitised beetle larvae reduces future recruitment of parasitoid population by killing them with beetle larvae before their emergence (Polis et al., 1989; Rosenheim, 1998; Fakhry et al., 2022). In a theoretical scenario of the absence of parasitoids, the generalist passeriform birds can still provide natural regulation of the beetle, but only during the outbreak, increasing the risk of canopy loss. Therefore, even if that theoretical equilibrium is stable, it is not desirable. Therefore, this research focuses on optimising intervention strategies that maintain coexistence equilibria and detect tipping points to shift to any other possible equilibria of the system.

Despite detailed empirical investigation into phenologies in this focal ecoepidemic system, the canopy state regime and adaptive stability remain poorly understood. Measuring resistance is challenging and may only provide a limited understanding of the current stage. For instance, it is empirically difficult to quantify the exact thresholds for plant chemical accumulation, required bird diversity, or parasitoid density needed to control the beetle population without risking extinction. In this study, we measure resistance as a loss from the expected outbreak intensity across seasonal and annual scales. We define resilience as the reciprocal of the rate of recovery in annual carrying capacity. However, following Van Meerbeek et al. (2021), we view ecological stability as a multidimensional property rather than a single measure. This perspective helps place our focal system within a broader stability framework that includes resistance, resilience, and the ability of the system to remain within the same domain of attraction after perturbation (Donohue et al., 2016; Hillebrand et al., 2018). Here, we use that framework only as a conceptual guide. The actual stability measures are defined using biologically relevant variables from our hybrid seasonal–annual model: low larval density and low cumulative defoliation indicate strong resistance during the outbreak season, whereas rapid return of annual carrying capacity and system state toward the reference annual equilibrium indicates resilience across years. In this way, Van Meerbeek et al. (2021) provide the general language of stability, while our model provides the system-specific quantities used to evaluate the objectives.

Using these definitions, we set three aims to advance the understanding of adaptive stability in this ecoepidemic system. This approach can support management planning under optimal control theory. O1. We identify the most probable stable equilibria of the system across all phenological scenarios. To meet this aim, we form a process-based model that captures phenologies and intervention strategies through parameters. We represent variability in phenologies via parameter space. We then search for all possible stable equilibria in this variable parameter space. Next, we test H1: under all variability in phenologies, the system is likely to converge to a desirable coexistence equilibrium, defined by the presence of a canopy, bird, and parasitoid that can prevent future beetle outbreaks if reintroduced. O2. We quantify the probability of the system shifting to an undesirable equilibrium from its current coexistence equilibrium, unless coexistence is the likeliest equilibrium across parameter space. In that case, we test H2: changes in certain phenologies, or parameter values, may serve as early warning signals of a shift to an undesirable equilibrium that still maintains stability. If so, we aim to classify them for each potential undesirable equilibrium. O3. We determine the minimal effort needed for three pest control strategies against the leaf beetle to prevent the system from shifting regimes across parameter space. Using optimal control, we aim to minimise the cost of effort per strategy. To guide

intervention planning, we test three sets of scenarios, ranging from the least costly to the most costly strategy combinations, against all variable phenologies.

2. Methodology

We use the structural property of the systems' well-studied ecological interactions to model the system using process-based modelling. Then we characterise the system's stability across parameter space in this study with emphasis on persistence conditions, equilibrium transitions, and comparative control performance. Therefore, we employ a four-stage analysis to characterise stability from the structural properties of the system within a unified seasonal–annual framework. First, we formulate a set of process-based within-season ordinary differential equation models and nest them within an annual recurrence process. Second, we identify equilibria as fixed points of the annual process and derive the basic reproduction rate of the parasitoid at the parasitoid-free fixed point using the next-generation matrix with modifications required for the hybrid seasonal-annual structure following van den Driessche & Watmough (2002). Third, we analyse the local stability of the annual fixed points through the Jacobian of the annual process, including computation of the Jacobian by means of variational (sensitivity) equations of the within-season flow, together with identification of bifurcation boundaries in parameter space (Kuznetsov, 2004). We define the parameter space based on eigenvalues of the annual-process Jacobian crossing the unit circle. Fourth, we formulate an optimal control problem and solve it numerically through Pontryagin's Maximum Principle and the forward–backward sweep method to test the stability of the system in light of optimal control theory (Lenhart & Workman, 2007). We then perform a systematic robustness evaluation across the full plausible parameter ranges in the numerical analyses (see Section 2.3) and compare the resulting optimal-control trajectories and stability landscapes across the alternative control scenarios introduced in the Introduction. We organise the methodology into three subsections: (1) model formulation, (2) structural stability analyses, and (3) numerical analyses. The model formulation includes an explicit mapping of nine phenological scenarios to model parameters (Table S1 in Annex 1). The structural stability analyses derive equilibrium conditions, the parasitoid invasion threshold, bifurcation boundaries, and optimal control solutions. The numerical analyses evaluate all stability and control outcomes across the full plausible parameter space defined by these phenological scenarios (Annex 1). Together, the three subsections address: (O1) identification of the most probable stable equilibria across phenological scenarios, (O2) quantification of regime shift probability and identification of parametric drivers of instability that may translate to candidate early warning signals, and (O3) determination of the minimal intervention effort under three pest-management scenarios.

2.1 Model formulation

The model comprises two coupled modules that operate at distinct temporal scales. Module 1 is a within-season eco-epidemic ordinary differential equation system that characterises the tri-trophic dynamics during the spring-summer larval vulnerability window of the leaf beetle. Module 2 is an annual discrete recurrence process that connects successive seasons by updating the beetle recruitment cohort, the parasitoid overwintering population, and the implicit alder foliage carrying capacity according to cumulative defoliation damage. All state variables are expressed per hectare of alluvial forest stand. Because resistance emerges within and across seasons, while resilience depends on annual canopy recovery and carrying-capacity return, only a two-timescale model can identify equilibria, regime shifts, and minimal interventions reliably.

Module 1: We consider four within-season state variables over continuous time $\tau \in [0, T]$, where T is the duration of the larval vulnerability window (baseline T = 55 days, range 45-70 days; univoltine phenology of the leaf beetle). We denote $S(\tau)$ as the density of susceptible (unparasitised) beetle larvae, $I(\tau)$ as the density of parasitised ("infected") beetle larvae, and $F(\tau)$ as the density of adult parasitoid flies. $D(\tau)$: cumulative defoliation damage. Population state variables are expressed per hectare of alluvial forest stand, whereas cumulative defoliation damage is represented as a dimensionless fractional quantity. We describe the within-season dynamics of susceptible beetle larvae by:

$$\frac{dS}{d\tau} = -\frac{\beta\, F\, S}{1+hS} - \frac{c_B\, B_t S}{1+a_B(S+I)} - \left(\mu_S + u_C(\tau)\right)S \text{ ... Equation 1}$$

Here $\frac{dS}{d\tau}$ is the rate of change of susceptible *beetle* larval density over the within-season time τ. The first term, $\frac{-\beta\, F\, S}{1+hS}$, represents the parasitoidism and follows a Holling type II functional response. Here, β denotes the parasitoid attack rate, and *h* denotes the parasitoid handling or saturation parameter; together they capture the fact that parasitoid attack increases with host availability but saturates at high larval density because oviposition is time-limited. The second term, $\frac{-c_B\, B_t S}{1+a_B(S+I)}$, represents predation by generalist passerine birds, also in Holling type II form, where total larval density (S+I) determines saturation of bird consumption. Here, c_B is the per-unit bird-pressure consumption rate, and B_t is an exogenous index of generalist passerine bird pressure in the focal forest stand rather than an explicit bird population state variable. We use an index because bird abundance and community composition are shaped by ecological processes operating beyond the focal stand and beyond the seasonal timescale of the beetle outbreak, whereas in this model, we require only their aggregate predation pressure on larvae (Monteagudo et al., 2023). The parameter a_B is the bird half-saturation or handling parameter and controls how quickly predation approaches saturation as larval density increases. It therefore reflects a saturating consumption process, not learned avoidance. The final term, $\left(\mu_S + u_C(\tau)\right)S$, represents the removal of susceptible larvae through baseline natural mortality and external intervention. Here, μ_S is a background/baseline mortality rate of larvae from causes other than parasitism and bird predation, and $u_C(\tau)$ is the time-dependent removal rate of larvae due to direct intervention during the within-season control process, and during the within-season intervention process.

We describe the within-season dynamics of parasitised ("infected") *beetle* larvae by

$$\frac{dI}{d\tau} = \frac{\beta FS}{1+hS} - \frac{c_B B_t I}{1+a_B(S+I)} - \left(\mu_I + \delta + u_C(\tau)\right)I \text{ ... Equation 2}$$

Here $\frac{dI}{d\tau}$ is the rate of infected beetle larvae, the first term, $\frac{\beta FS}{1+hS}$, represents the conversion of susceptible larvae into the parasitised class through oviposition by parasitoid, and thus forms the eco-epidemic core of the system. The second term, $\frac{c_B B_t I}{1+a_B(S+I)}$, represents predation on parasitised larvae by generalist passerine birds in the same Holling type II form used for susceptible larvae, with total larval density (S+I) determining saturation. The final term, $\left(\mu_I + \delta + u_C(\tau)\right)I$ represents removal of parasitised larvae through background mortality, parasitoid emergence, and external intervention. Here, μ_I is the background mortality rate of parasitised larvae, which we assume to be at least slightly higher than μ_S to reflect the additional physiological burden associated with parasitoid development within the host. The parameter δ denotes mortality due to the emergence of the adult parasitoid from the host and therefore represents the delayed lethal effect of parasitism; its value is linked to reported tachinid developmental timing and host emergence intervals (Hawkins, 1994; Stireman et al., 2006).

$$\frac{dF}{d\tau} = \eta\, \delta\, I - \mu_F F + u_P(\tau) \text{... Equation 3}$$

$\frac{dF}{d\tau}$ is the rate of change in the population density of the parasitoid. Here $\eta\, \delta\, I$ represents the recruitment of new adult parasitoids: each parasitised larva that undergoes parasitoid-induced death (at rate δ) yields η new adult flies, where η is mathematically the conversion efficiency. For the solitary endoparasitoid, $\eta \leq 1$ (one parasitoid per host); $\eta < 1$ accounts for parasitoid mortality during development within the host. The second term is the natural mortality of adult parasitoid flies. The third term is the parasitoid augmentation control, where $u_{P(\tau)}$, representing management actions that increase parasitoid density (e.g., intentional release of parasitoids in the forests).

$$\frac{dD}{d\tau} = \kappa(S+I), D(0) = 0, D(T) \equiv D_t \text{ ... Equation 4}$$

$\frac{dD}{d\tau}$ represents the accumulation of the total defoliation damage D inflicted by both susceptible and parasitised larvae over the season. The parameter κ converts larval density into a fractional defoliation rate. Ecologically, it represents the foraging rate of larvae on the leaves of Black Adler. The terminal

value $D(T) = D_t$ represents the end of season defoliation and is passed to Module 2, which determines the reduction in carrying capacity of the beetle for the following year. Both S and I contribute to damage because parasitised larvae continue to feed until the parasitoid completes development.

Module 2: We model the temporal map of the annual recurrence that links successive within-season modules in this second module. It updates beetle recruitment, parasitoid carryover, and the implicit alder carrying capacity. The key ecological process captured here is the resilience mechanism: the deciduous *Alnus glutinosa* regenerates its entire canopy each spring from new foliage, restoring the carrying capacity for beetle larvae, but only to the extent permitted by the tree's vigour, which is penalised by cumulative defoliation damage from the previous season. This module consists of 5 main equations and 2 additional equations for optimal control and threshold identification for regime shift during stability analyses.

$S(0) = \frac{R_B A_t}{1+\frac{A_t}{K_t}}, I(0) = 0, F(0) = F_t$...Equation 5

Here $S(0)$ The initial susceptible larval density at the start of each season is determined by the Beverton-Holt recruitment function (Beverton and Holt, 1957), which limits the recruiting cohortss A_t to the current carrying capacity of beetles (K_t) through density-dependent competition for foliage. R_B is here annual beetle reproduction rate calculated from its fecundity. No parasitised larvae exist at season onset ($I(0) = 0$) because parasitoid activity begins only when adult *M. mutabilis* encounter susceptible host larvae within the season. The initial parasitoid density equals the overwintering carryover F_t.

$A_{t+1} = \sigma_A S(T), F_{t+1} = \sigma_F F(T)$ ...Equation 6

At the end of each within-season window, the surviving susceptible larvae S(T) are mapped to the recruiting cohort for the next year (A_{t+1})through an overwinter survival fraction σ_A . This term mathematically represents a single annual survival parameter that captures all the overwintering survivals: pupation in soil, cold-season mortality, adult emergence, mating, and oviposition. Similarly, σ_F represents end-of-season adult parasitoid density to next year's initial density, accounting for tachinid puparia overwinter survival.

$B_t = B_t^{\text{index}}\big(1 + \xi\, u_B(t)\big)$... Equation 7

Here B_t is the effective bird pressure in each year is the product of the baseline bird index B_t^{index} (obtained from PECBMS monitoring data or equivalent woodland bird surveys) and a management multiplier $\big(1 + \xi\, u_B(t)\big)$. The control variable $u_{B(t)}$ represents annual bird-habitat enhancement actions (e.g., nest-box provisioning, structural habitat improvements, reduced disturbance during the breeding season). The parameter ξ scales the management effort to its effect on local bird density, i.e. the increment or change of bird index as a response to management effort $u_B(t)$.

$K_{t+1} = K_0 \exp(-\phi D_t)$... Equation 8

K_{t+1} is the carrying capacity of consecutive years that captures the core resilience of the system due to inter-annual feedback from cumulative defoliation. K_0 is the undamaged baseline carrying capacity (corresponding to a fully recovered canopy from a healthy *Alnus glutinosa* stand). $\text{Exp}(-\phi D_t)$ is a penalty that depends upon the accumulated damage D_t. The penalty reduces proportionally to next year's foliage quality and quantity, as per the empirical findings of Ferretti et al. (2021) and Dolch and Tscharntke (2000): broadleaf tree growth decreases by approximately 0.9% per 1% increase in defoliation, and with the induced chemical resistance and persists through the growing season (~133 days) and reduces subsequent beetle herbivory in the following year.

Finally, to analyse the optimal control of the system, we define an objective functional that minimises the total ecological and management cost of beetle damage and intervention over a multi-year planning horizon of N years, where each year contains one within-season window of duration T.

$J = \sum_t^N \left[\int_0^T \left(\omega_D D(\tau) + \omega_S S(\tau) + \frac{c_P}{2} u_P(\tau)^2 + \frac{c_C}{2} u_C(\tau)^2 \right) d\tau + \frac{c_{B,cost}}{2} u_B(t)^2 + \omega_T D_t \right]$... Equation 9, subject to $0 \leq u_P(\tau) \leq u_P, max, u_C(\tau) \leq u_C, max, 0 \leq u_B(t) \leq u_{B,max}, max$...Equation 10

Here ω_D, ω_S, and ω_T are the weights that penalise cumulative damage, current larval density, and terminal damage respectively. Because D(τ) accumulates over the season, the running penalty $\omega_D D(\tau)$ assigns greater weight to damage sustained later in the season, reflecting the increasing physiological cost of prolonged defoliation. Ecologically, these are the feedback to the carrying capacities for the beetle larvae through accumulation of canopy loss, accumulation of susceptibility to the infection from parasitoid, and accumulation of end-of-season defoliation that reduces the chances of overwintering of the beetles.$c_P, c_C and\ c_{B,cost}$ are the unit costs for the three respective processes: Unit cost of parasitoid attack on larvae to other forest economy, Unit cost of targeted removal of larvae, Unit cost of bird-habitat conservation. However, for the simplicity of the model, we consider these costs as a fraction of the forest management budget and, therefore, dimensionless.

Then we also use a function to capture the central quantity for the stability analysis as the parasitoid invasion regime threshold R_P. In the hybrid seasonal-annual architecture of this model, R_P is defined operationally as the dominant eigenvalue (spectral radius) of the parasitoid update linearised around the parasitoid-free fixed point of the annual map, rather than from the within-season ODE equilibria alone. This definition follows the next-generation approach for compartmental models (van den Driessche and Watmough, 2002) adapted to the hybrid structure via the Poincaré (stroboscopic) map framework (Kuznetsov, 2004). At the parasitoid-free annual fixed point $x_1 * = (A^*, 0, K^*)$, the parasitoid component F is absent, and the beetle–foliage subsystem is at its resident equilibrium. The capacity of a small parasitoid introduction to grow across one full seasonal cycle is:$R_P \equiv \frac{\partial F_{t+1}}{\partial F_t}|_{x_1^*} = \sigma_F \frac{\partial F(T)}{\partial F(0)}|_{S(0)=\hat{S}, I(0)=0, F(0)=0}$ ... Equation 11

Here $\frac{\partial F(T)}{\partial F(0)}$ is the partial derivative obtained from the variational (sensitivity) equations of within season ODE. When the within-season dynamics are evaluated at constant S ≈ S̄ and constant bird pressure B_t. This rigorous definition reduces to the interpretable approximation: $R_P \approx \frac{\eta\, \delta\, \beta\, \hat{S}}{(1+h\hat{S}) \cdot \mu_F \cdot \left(\mu_I + \delta + u_C + \frac{c_B B_t}{1+a_B \hat{S}} \right)}$.

Here $R_P < 1$ denotes parasitoid dies out eventually in a regime, and the equilibrium would be parasite-free. $R_P > 1$ means the parasitoid can invade and persist, and it represents the present endemic parasitism regime where biological control is effective. Therefore $R_P = 1$ is the transcritical bifurcation boundary, where we can observe a regime shift. The $R_P = 1$ is searched by controlling the following model parameters: $\beta, \eta, \mu_F, c_B B_t, u_C$. The rest of the model parameters are obtained from the literature and listed in Annex 1.

2.2 Phenological scenarios and parameter mapping

The nine phenological scenarios described in the Introduction are represented in the model through variation of specific parameters. Table S1 maps each scenario to its climate driver, the affected model parameter(s), the baseline value, and the perturbed range. Each range is justified by published literature or biologically constrained priors (Annex 1). In the numerical analyses, the effect of each scenario is first assessed individually (one-at-a-time variation holding other parameters at baseline) to isolate the contribution of each phenology, and then jointly through full Latin Hypercube Sampling across all parameter ranges to capture interactions among scenarios.

2.3 Stability Analyses

The long-term stability of this system is determined by the annual recurrence map rather than by the within-season ordinary differential equation (ODE) considered in isolation. The within-season ODE (Equations 1-4) characterises loss and accumulation processes over a finite interval τ ∈ [0, T]. Setting $\frac{dS}{d\tau} = \frac{dI}{d\tau} = \frac{dF}{d\tau} = 0$ produces within-season pseudo-equilibria, which do not represent ecologically relevant steady states, since the system is reset annually through recruitment, overwinter survival, and foliage recovery equations (5-8). Multi-year dynamics, including outbreak recurrence, parasitoid

persistence, and tipping risk, are governed by the annual map. The stability analysis is analytically well-posed but does not admit a closed-form solution. Specifically, the annual map G and its structure, existence conditions for fixed points, the Jacobian J_G of the annual map (derived via the chain rule and variational equations), the parasitoid invasion threshold R_P as the dominant eigenvalue of J_G, local stability criteria (spectral radius less than one), and bifurcation conditions (eigenvalue crossing the unit circle) are all derived analytically. In contrast, the seasonal flow Φ_T and its derivatives are evaluated numerically, as the Holling Type II functional responses render the within-season ODE nonlinear and analytically intractable. This approach, which combines analytic stability conditions with numerical evaluation of the seasonal flow, is standard and mathematically rigorous for hybrid eco-epidemic models (Kot, 2001; Kuznetsov, 2004). The analysis proceeds through a 10-step framework: 1. Definition the annual state and the seasonal flow operator, 2. Construction of the annual recurrence map 3. Identification of fixed points (annual equilibria) of the map 4. Assessing local stability of annual fixed points via the Jacobian of the annual map 5. Derivation of the seasonal flow via variational (sensitivity) equations. 6. Calculation of Parasitoid invasion threshold from the annual map 7. Identification of tipping points as bifurcation boundaries of the annual map. 8. Assessing Optimal control via Pontryagin's Maximum Principle and the forward-backwards sweep method 9. Searching Tipping under optimal control 10. Assessing robustness under parameter uncertainty. The detailed calculation of these steps is in Annex 2, and has been generated via Mathematica, MatchCad, and manual calculation.

The stability analysis addresses the three study objectives through a stepwise analytical structure. For Objective 1 (O1), all ecologically distinct fixed points of the annual map are identified (Step 3). Their local stability is classified using the Jacobian eigenvalues (Step 4). The parasitoid invasion threshold R_P is then computed (Step 6). The analysis scans the full parameter space to determine the frequency of each equilibrium type across phenological scenarios. The equilibrium with the highest frequency is designated as the most probable equilibrium. Objective 2 (O2) is addressed by tracking eigenvalue magnitudes and bifurcation boundaries across parameter space (Step 7). The regime shift probability is computed as the fraction of sampled parameter sets converging to an alternative equilibrium. Parameters are ranked by their influence on bifurcation proximity using the PRCC of the dominant eigenvalue magnitude (Step 10). Parameters identified as drivers of instability become candidate early warning signals if the corresponding phenology is empirically observable and its variation matches the destabilising parameter direction. Objective 3 (O3) is addressed by optimal control analysis (Step 8) and a three-scenario comparison (Step 9). This compares resistance, resilience, stability, and cost outcomes across parasitoid-only, combined parasitoid-and-bird, and full integrated control, applying a lexicographic decision rule: feasibility first, then minimum cost.

2.4 Definitions of resistance, resilience, and tipping during stability analyses

In this study, the framework of Van Meerbeek et al. (2021) is used to structure the interpretation of stability, not to replace the model-specific metrics required by the present eco-epidemic system. Their synthesis treats ecological stability as a set of complementary properties that describe how a system responds to perturbation relative to a reference condition and its domain of attraction. We therefore map those general properties onto the quantities generated by our annual map. Resistance is represented by suppression of outbreak intensity, measured through equilibrium defoliation and peak larval load; resilience is represented by the rate of return to the annual equilibrium, measured through the dominant eigenvalue magnitude and return time; and tipping is evaluated from loss of local stability and from ecologically relevant functional thresholds. The parasitoid invasion threshold $R_P$is included as a complementary regime-structure metric for Objective 2, rather than as a direct Van Meerbeek stability component.

We operationalise the first two components for the hybrid seasonal-annual model as follows. All metrics are standardised against a reference state and bounded between −1 and 1, enabling comparison across phenological scenarios. We apply these standardised metrics to model-simulated trajectories from the annual recurrence map, treating the map output as the system time series from which the reference value and perturbation response are extracted. Reference state. For each system variable, the reference state S_R is defined as its value at the annual fixed point $x^* = (A^*, F^*, K^*)$. Specifically, resistance is assessed using cumulative defoliation D as the system variable, with $S_R = D^*$

(equilibrium defoliation); resilience is assessed using carrying capacity K as the system variable, with $S_R = K^*$ (equilibrium carrying capacity). Different system variables may have different reference states, consistent with the multidimensional treatment of stability (Donohue et al., 2016; Hillebrand et al., 2018). Resistance is the capacity of the system to withstand change during a perturbation (Van Meerbeek et al., 2021). It is measured by the standardised index: $Resistance = 1 - \frac{2|S_R - S_X|}{S_R + |S_R - S_X|}$(R1), where S_X is the value of the system variable at maximum deviation following a standardised pulse perturbation to the annual map (displacement of A_t by ± δ% of A^*, with δ = 20% as the baseline perturbation magnitude). A resistance value of 1 indicates no deviation from the reference; 0 indicates deviation equal to the reference value. As complementary indicators, we also report the equilibrium cumulative defoliation D* and the peak within-season total larval density $max_\tau[S(\tau) + I(\tau)]$ along the seasonal trajectory starting from fixed-point initial conditions. Resilience is the rate at which the system returns to its reference state after a perturbation (Van Meerbeek et al., 2021; Pimm, 1984). It is measured by the standardised index: $Resilience = \frac{2|S_R - S_0|}{|S_R - S_0| + |S_R - S_Y|} - 1$ (R2), where S_0 is the value of the system variable at maximum deviation from the reference (i.e., $S_0 = S_X$ from the resistance measurement), and S_Y is the value after Y annual iterations following the perturbation. We set Y = 5 annual cycles as the standardised recovery window, corresponding to a management-relevant planning horizon. A resilience value of 1 indicates full recovery within Y iterations; 0 indicates no net recovery. As complementary indicators, we report the dominant eigenvalue magnitude $\rho^* = max_i \left|\lambda_i\left(J_G(x^*)\right)\right|$ of the annual-map Jacobian at the fixed point and the characteristic return time $T_{return} = -\frac{1}{ln(\rho^*)}$, which captures asymptotic (linearised) resilience. Eigenvalue-based and trajectory-based resilience provide complementary information: the former reflects the local linear return rate, while the latter captures nonlinear dynamics far from the fixed point (Arnoldi et al., 2016; Medeiros et al., 2021).

The two metric sets differ in their mathematical domain of validity. The eigenvalue-based indicators ρ* and $T_{return} = -\frac{1}{\ln(\rho*)}$ are properties of the Jacobian J_G evaluated at the fixed point x*; they characterise the linearised return rate and are therefore valid only for infinitesimally small perturbations within the immediate neighbourhood of x*. In contrast, R1 and R2 are computed from the full nonlinear annual map iterated from a finite-amplitude perturbation (±20% displacement of A_t). Because the Holling Type II functional responses in Equations 1-2 introduce curvature that the Jacobian cannot capture, the nonlinear trajectory may deviate substantially from the linearised prediction when the perturbation is large or when the system state approaches a basin boundary. Consequently, ρ* and R2 need not rank parameter scenarios identically: a fixed point with ρ* close to unity (slow linearised return) may nonetheless exhibit adequate finite-time recovery if the nonlinear restoring forces are stronger than the linearisation predicts, while a fixed point with small ρ* (fast linearised return) may yield low R2 if the perturbation places the trajectory near a separatrix between basins of attraction. Including both metric sets, therefore, ensures that stability assessment is not an artefact of the linearisation domain.

In a system with alternative stable states, a perturbation that pushes the system into a different basin of attraction will yield recovery near 0 or negative; a perturbation from which the system returns to x* yields recovery = 1. Latitude is the maximum perturbation magnitude the system can absorb before losing its ability to recover (Walker et al., 2004; Van Meerbeek et al., 2021). It is computed by progressively increasing the perturbation magnitude δ until the annual map fails to return to the original equilibrium within a specified horizon (50 annual iterations). The critical δ at which the system transitions to an alternative attractor defines the latitude boundary. This is related to the basin stability concept of Menck et al. (2013). Adaptive stability. We define adaptive stability as the capacity of the system to maintain ecological function through the joint action of within-season resistance and annual resilience, where the balance between resistance and resilience adapts to phenological conditions. Unlike engineering resilience (a single return-rate measure; Pimm, 1984), ecological resilience (a basin-of-attraction measure; Holling, 1973), or adaptive capacity (a social-ecological management concept; Folke et al., 2005), adaptive stability captures the multi-timescale interplay in which within-season resistance to outbreak damage and annual recovery of canopy condition jointly determine whether the system persists in a functional regime. The term is novel to this study; it is operationalised as the joint

profile of resistance (R1) and resilience (R2) across phenological scenarios, rather than as a single scalar index. Three levels of stability loss. We distinguish three levels of stability loss that may or may not coincide (Van Meerbeek et al., 2021; Scheffer et al., 2001): (i) Loss of local asymptotic stability: the dominant eigenvalue magnitude ρ* crosses unity, and the annual fixed point loses its local attracting property. (ii) Functional threshold crossing: the system variable enters a functionally unacceptable range—specifically, the long-run mean defoliation exceeds $D_{crit} = 0.5$ (corresponding to 50% seasonal defoliation; Ferretti et al., 2021), or the carrying capacity K_t falls below $K_{min} = 0.5K_0$ for three or more consecutive years, indicating persistent canopy degradation, while potentially remaining in the same dynamic regime. (iii) Regime shift: the system transitions to a qualitatively different equilibrium (e.g., from coexistence to a parasitoid-free state), meaning the long-term dynamics converge to a structurally different attractor. The dynamical tipping boundary is the critical parameter value p_c at which $max_i\,|\lambda_i(p_c)| = 1$ (Equation 19). The functional tipping criterion is evaluated relative to the dynamical boundary to determine whether ecological collapse precedes or follows the loss of mathematical stability.

Regime shift probability and parametric drivers of instability for O2. The probability of regime shift is defined as the fraction of N LHS-sampled parameter sets (Step 10) for which the annual map converges to an equilibrium qualitatively different from the coexistence fixed point x_2^*. Formally, $P_{shift} = \frac{n_{alt}}{N}$, where n_{alt} is the number of sampled parameter sets yielding a parasitoid-free or trivial equilibrium. This probability is conditional on the assumed uniform distributions over the parameter ranges in Annex 1 and is reported with 95% bootstrap confidence intervals. Parametric drivers of instability are identified by computing PRCC between each model parameter and the dominant eigenvalue magnitude ρ* across the LHS sample. Parameters with $|PRCC| > 0.25$ that drive ρ* toward unity are classified as parametric drivers of instability. These parametric drivers can be translated to candidate early warning signals when the phenological process providing ecological meaning to the parameter is empirically observable, and its direction of variation under climatic change matches the destabilising direction identified by the model. For example, if the model identifies overwinter beetle survival σ_A as a parametric driver whose increase destabilises coexistence, and if milder winters observably increase σ_A Then, declining winter severity constitutes a candidate early warning signal for regime shift. As a focused analysis within O2, we compute R_P across the full range of the bird pressure index $B_t \in [0.5, 2.0]$ at the calibrated baseline, and across the (β, B_t) parameter plane, to map the transcritical bifurcation boundary separating the coexistence and parasitoid-free regimes.

2.5 Numerical analyses

The numerical analyses are conducted across all nine phenological scenarios (Table S1), first by one-at-a-time parameter variation to isolate the effect of each scenario, and then by joint Latin Hypercube Sampling (JLHS) across all parameter ranges to capture interactions. This design enables the computation of equilibrium frequencies (O1), regime shift probabilities (O2), and strategy performance rankings (O3) across the full phenological parameter space.

The absence of long-term, site-specific time series for Beetle outbreak density, M. mutabilis parasitism prevalence, and passerine kill rates on this beetle challenges only numerical simulation-based analyses. Therefore, we focus on the system characteristic through mapping the structural stability landscape of the system and calibration for parameter ranges per regime. This technique allows us to identify parameter combinations and control allocations that support a stable low-damage equilibria: those that induce tipping, and the response of the parasitoid invasion threshold R_P to intraguild predation and control effort. However, this analysis therefore restricts us from aiming to predict site-specific outbreak timing or magnitude without site-specific time series data. Stability properties, including fixed-point existence, eigenvalue conditions, and bifurcation type, are determined by the structure of the annual map and the qualitative nature of the functional responses in our simulation for a range of plausible values based on stability analyses in annex 2 and known parameter values in annex 1. We follow this parameter range for simulation in the standardised protocol in theoretical eco-epidemic modelling, as it is in Chattopadhyay and Arino, 1999; Hethcote et al., 2004.

During numerical analyses, parameters were divided into four classes: biological baseline parameters, calibrated structural parameters, bounded management controls, and objective-function weights. Biological parameters were assigned from empirical studies, cross-system proxies, or biologically constrained priors as summarised in Annex 1. Structural parameters lacking direct estimates but required for scale consistency or threshold behaviour, including β, c_B, and κ, were calibrated so that the model reproduced plausible baseline parasitism, daily bird predation, and proportional defoliation. The management variables $u_P(\tau)$, $u_C(\tau)$, and $u_B(t)$were not treated as ecological constants, but as bounded decision variables with baseline value zero. The weights in the objective functional were treated as scenario-dependent management preferences rather than empirically fixed ecological parameters.

To address the lack of precise parameter estimates, the analysis employs two complementary safeguards. First, all uncertain parameters are represented as intervals, as specified in the parameter tables of Annex 1, with empirically justified ranges. As a result, stability and control outcomes are reported as robust regions within parameter space rather than as point estimates. Conclusions are accepted only if they are valid across the entire range of each uncertain parameter. Second, structural robustness is assessed by replacing the baseline functional forms with ecologically plausible alternatives: Beverton-Holt versus Ricker recruitment in Equation (5), Holling Type II versus linear functional response for bird predation, and S → I compartmental parasitism versus direct parasitoid-induced mortality. If qualitative stability conclusions—specifically, which equilibrium is most probable (O1), whether the regime shift probability exceeds a policy-relevant threshold (O2), and which strategy is minimal-effort feasible (O3) remain consistent under both parametric and structural perturbations, these results are interpreted as intrinsic properties of the tri-trophic interaction architecture rather than artefacts of specific modelling assumptions.

For each equilibrium and each control scenario, we compute the resistance metrics $D^*$and $max_\tau[S(\tau) + I(\tau)]$, the resilience metrics $\rho^*$and $T_{\text{return}} = -\frac{1}{\ln(\rho^*)}$, and cumulative control cost J where relevant. A perturbation-recovery time obtained by iterating the annual map from a standardised displacement of the fixed point, so that the relative performance of parasitoid-only and combined control strategies can be compared on resistance, resilience, and cost grounds.

3. Results

The results of this study are grouped by each objective. Each subsection from 3.1 to 3.3 reports the metrics that directly test the corresponding hypothesis.

3.1 Meeting objective 1

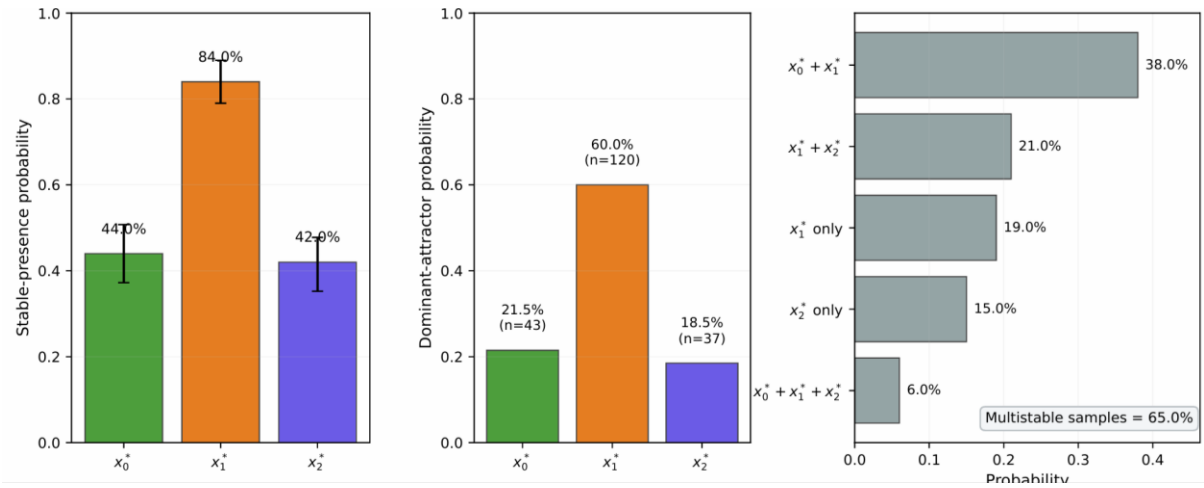

Figure 1: Equilibrium landscape across the phenological parameter space. x_0^*, x_1^*, $x_2^*$are canopy-only, parasitoid-free, and coexistence equilibria. No degraded equilibria (i.e., carrying capacity below half the threshold value) were found. (A) Baseline fixed points of the annual map at calibrated parameters; bar length indicates equilibrium beetle density A*, with spectral radius ρ* and stability status annotated.

(B) Fraction of the Latin Hypercube–sampled parameter space in which each equilibrium class exists as a locally stable fixed point; bars can exceed unity in sum due to multistability. (C) Dominant-attractor frequency: the fraction of sampled parameter sets for which the annual map trajectory converges to each class from the coexistence reference initial condition; multistability rate annotated.

Figure 1 shows 3 outputs from stability analyses: Annual equilibria identified as fixed points of the yearly recurrence map (Step 3), their local stability classified from the Jacobian eigenvalues of that annual map (Step 4), and the frequency of each equilibrium occurs across Latin Hypercube Sampling under parameter space (Step 10). Figure 1A reports the stable-presence probability, i.e., the fraction of sampled parameter sets for which a given equilibrium exists as a locally stable fixed point of the annual map. In this sense, the parasitoid-free equilibrium canopy-only is the most frequent stable regime across parameter space, whereas the coexistence equilibrium parasitoid-free is the least frequent stable regime found across the parameter space. The stable presence probability of the parasitoid-free equilibria is closest to the coexistence equilibria. Therefore, under continuous variable phenology, the phenological mismatch can lead to a parasitoid-free equilibrium where the beetle outbreak is regulated by birds, and eventually the system is likely to jump into a stable equilibrium where the beetle is also absent. This result indicates rejection of the H1: under all variability in phenologies, the system is likely to converge to a desirable coexistence equilibrium, defined by the presence of a canopy, bird, and parasitoid that can prevent future beetle outbreaks if reintroduced.

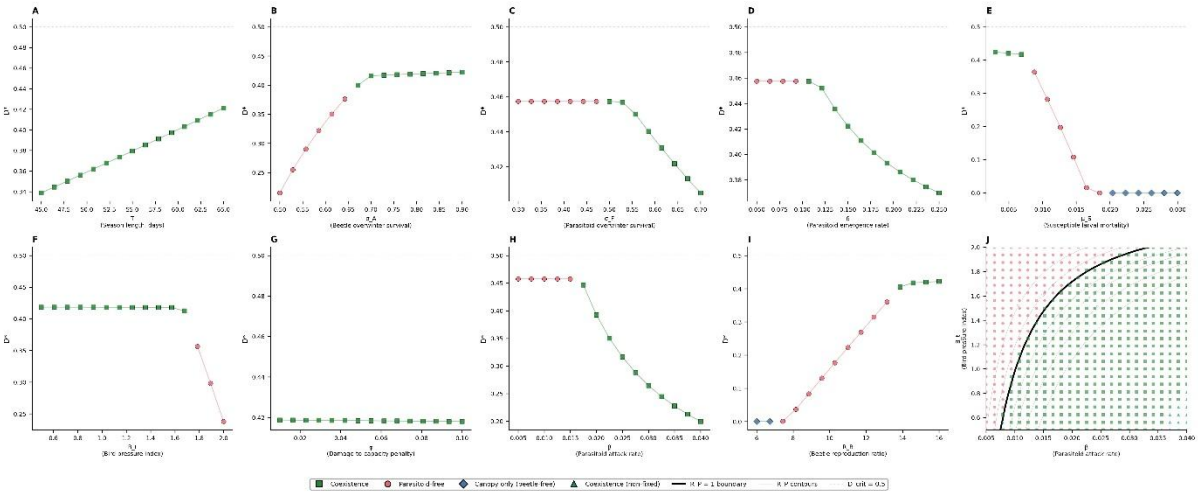

Figure 2 Equilibrium transition boundaries across all nine phenological scenarios and the pairwise $\beta \times B_t$ parameter plane. Panels A-I show equilibrium damage D^* at the stable fixed point of the annual map as each phenological parameter is varied individually across its plausible range (one-at-a-time scans; all other parameters held at the calibrated baseline). Green squares: coexistence equilibria; pink circles: parasitoid-free equilibria; blue diamonds: canopy-only (beetle-free) equilibria; teal triangles: coexistence non-fixed-point attractors. Connecting lines trace the equilibrium branch within each class. The horizontal dotted line marks the functional damage threshold $D_{crit} = 0.5$. (A) Season duration T. (B) Beetle overwinter survival σ_A . (C) Parasitoid overwinter survival σ_F. (D) Parasitoid emergence rate δ. (E) Susceptible larval mortality μ_S. (F) Passerine bird pressure index B_t. (G) Damage-to-capacity penalty φ. (H) Parasitoid attack rate β. (I) Beetle annual reproduction ratio R_B. Panel J shows the pairwise phase plane in the $\beta \times B_t$ space; grey contours indicate R_P values and the bold black curve marks the $R_P = 1$ transcritical bifurcation boundary separating the coexistence regime (lower-right) from the parasitoid-free regime (upper-left). The diagonal orientation of this boundary reflects the intraguild predation mechanism: increasing bird pressure progressively raises the parasitoid attack rate required for parasitoid invasion.

Following the rejection of H1, Figure 2 presents the predicted stable fixed point of the annual map and the equilibrium damage D^* at the coexistence equilibrium. The results are displayed as a function of individual parameter variability, excluding parasitoid invasion risk as a stability driver. This analysis demonstrates the persistence of coexistence across different phenology variability ranges. Although

most phenologies and associated parameters covary in natural systems, examining individual variation reveals the system's maximum potential. Among nine one-at-a-time parameter scans, seven parameters altered the equilibrium class. Bird pressure (B_t) and the foliage-memory penalty (ϕ) did not induce regime shifts within the scanned range. Season length (T) influenced defoliation damage but did not transition the system to alternative equilibria within the examined range. Nevertheless, because defoliation damage responds rapidly to changes in T, a regime shift may occur over a longer temporal scale. Notably, reducing the parasitoid attack rate β, parasitoid overwinter survival σ_F, or the parasitoid emergence rate δ each caused a transition from coexistence to herbivore dominance, whereas increasing the beetle reproduction ratio R_B facilitated parasitoid invasion, directly affecting system stability and the resulting equilibrium class.

The most pronounced regime shift is driven by the parasitoid attack rate β. As β increases, system outcomes transition from the absence of parasitoids to coexistence, accompanied by a rapid reduction in equilibrium damage. The beetle reproduction ratio R_B delineates three distinct regimes: at low R_B, beetles are unable to persist and only trees remain; at intermediate R_B, beetles persist without parasitoids and inflict canopy damage; at high R_B, host density supports parasitoid establishment, resulting in stable coexistence. Increasing susceptible larval mortality μ_S initially disrupts parasitoid persistence by limiting host availability, and at higher values, beetles are eliminated, restoring the system to an intact canopy. Thresholds for regime shifts are also observed in beetle overwinter survival σ_A and parasitoid survival σ_F, with coexistence possible only above critical values. For the parasitoid emergence rate δ, coexistence is maintained only above a specific threshold; otherwise, the system remains parasitoid-free. Increasing bird pressure Bt does not create regime boundaries but alters equilibrium beetle and damage levels. Birds suppress beetle populations, reducing damage, and remove parasitised larvae, thereby weakening parasitoids. Consequently, their net effect modifies quantitative but not qualitative equilibrium classes within the scanned range. The phase plane illustrates how attack rate and bird predation influence equilibria (Figure 2J). Higher bird predation increases the attack rate required for coexistence, which predominantly occurs at high attack rates and low-to-moderate predation (71% of the grid). The parasitoid-free regime is observed at low attack rates or high predation (27% of the grid). Coexistence is achieved when parasitoid attack rate, winter survival, emergence rate, beetle reproduction and survival are high, larval mortality is low, and passerine predation is moderate, assuming other phenologies are held constant.

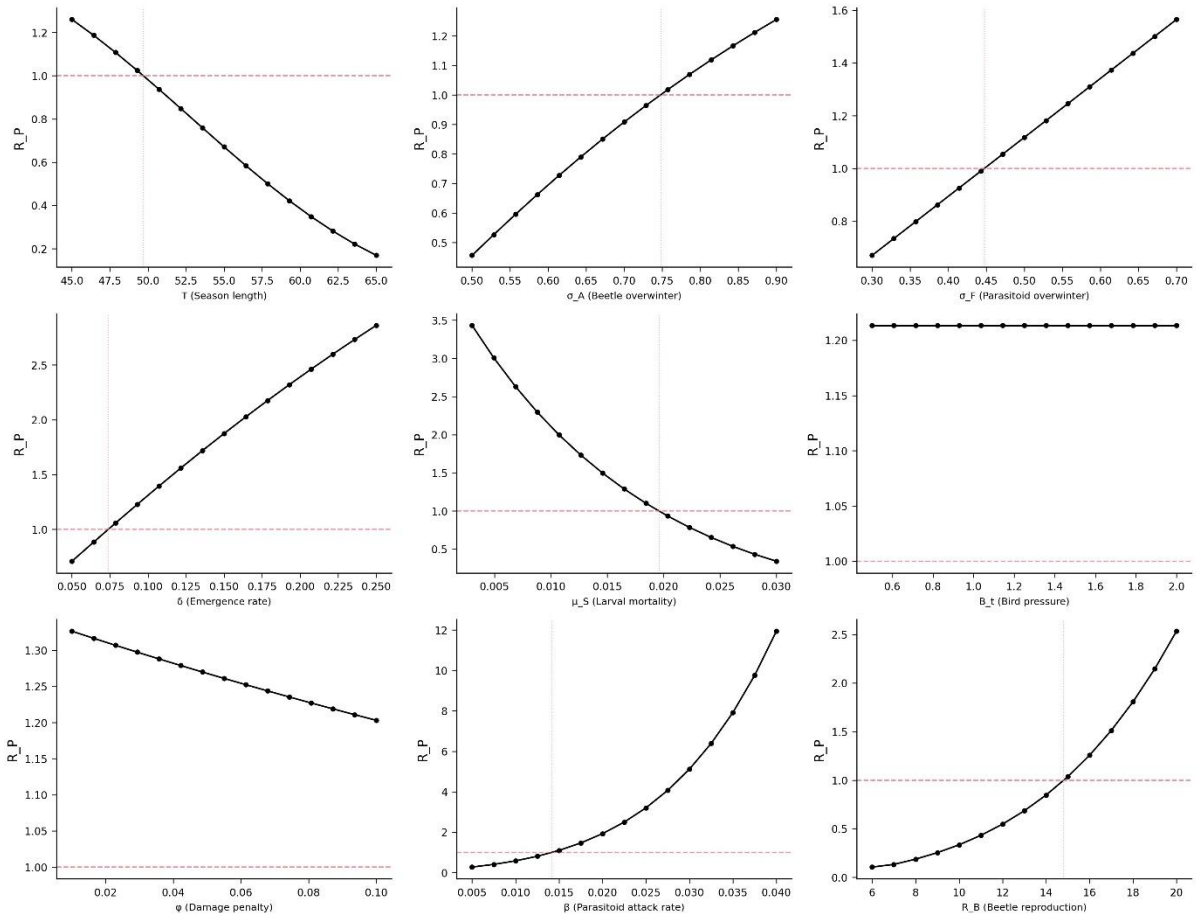

Figure 3: Parasitoid invasion threshold R_P as a function of each phenological parameter. Each panel shows R_P evaluated at the parasitoid-free fixed point of the annual map while varying one parameter across its plausible range (Table S1), with all other parameters held at the calibrated baseline. The dashed red line marks $R_P = 1$. Above this boundary, the parasitoid can invade and persist (coexistence regime); below it, the parasitoid declines to extinction (parasitoid-free regime). Vertical dotted lines indicate the approximate parameter value at which the threshold crossing occurs.

Figure 3 differs mathematically from Figure 2 by evaluating the parasitoid invasion threshold, R_P, at the parasitoid-free fixed point of the annual map, in contrast to the durability of the coexistence equilibrium. R_P measures the annual growth potential of a rare parasitoid at the boundary equilibrium. If R_P exceeds 1, parasitoids can increase from rarity and drive the system toward coexistence; if R_P is less than 1, they fail to establish, and the parasitoid-free state persists. The line $R_P = 1$ marks the invasion boundary between these regimes. This threshold measures how many new adult parasitoids are produced for each added per year, calculated at the parasitoid-free equilibrium of the annual map across other parameter space (Step 6). Since H1 is rejected, the parasitoid-free state is likely. Thus, R_P is crucial for O2. If $R_P > 1$, a few parasitoids can spread and push the system to coexistence. If R_P < 1, they die out, and the system remains parasitoid-free. The $R_P = 1$ line marks where the main shift between these two states occurs. Seven of nine parameters cross the $R_P = 1$ threshold. Raising the bird pressure index B_t quickly lowers R_P and crosses the threshold. Increasing the susceptible larval death rate μ_S also lowers R_P and crosses the threshold. Conversely, increasing the parasitoid attack rate β raises R_P and crosses the threshold. Higher R_B increases R_P with persistence around the threshold value of R_B from Figure 2. Beetle overwinter survival σ_A (Figure 3B), parasitoid overwinter survival σ_F (Figure 3C) and increasing parasitoid emergence rate δ (Figure 3D) each raises R_P. Each threshold crossing shows the parameter value needed for persistence. Season duration T (Figure 3A) increases R_P, so longer seasons lower but do not eliminate parasitoid persistence at baseline. The defoliation damages memory-based vernal growth φ (Figure 3G), which defines carrying capacity, slightly

decreases R_P. This shows that foliage-memory feedback has minimal impact on parasitoid invasion. Therefore, Figure 3 reveals that moving from coexisting equilibria by decreasing R_P results from a lower attack rate, reduced beetle or parasitoid survival, or slower emergence.

3.2 Meeting objective 2

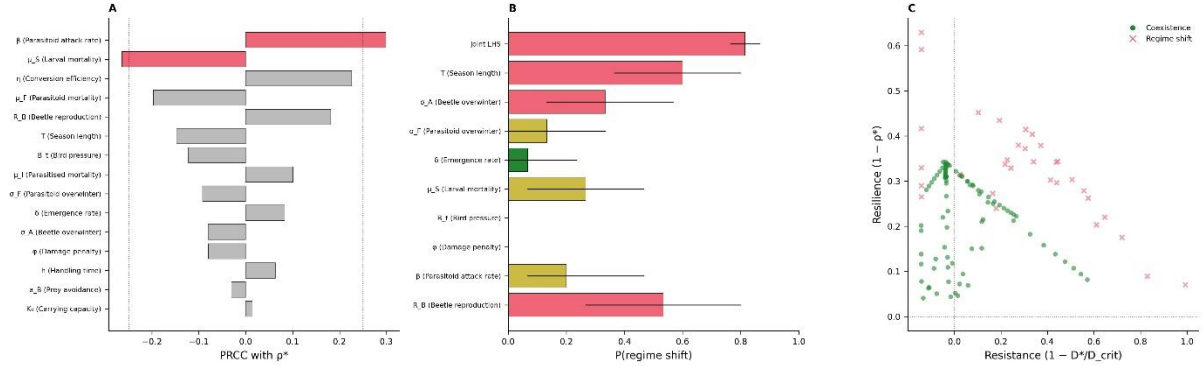

Figure 4: Early-warning summary: parametric drivers, regime-shift probability, and adaptive-stability reference frame. (A) Partial rank correlation coefficients (PRCC) between each model parameter and the dominant eigenvalue magnitude ρ across the Latin Hypercube sample; red bars indicate |PRCC| > 0.25 (warning candidates). (B) Regime-shift probability per phenological scenario (one-at-a-time variation) and for the joint Latin Hypercube sample; horizontal bars show 95% bootstrap confidence intervals; colour indicates risk level. (C) All one-at-a-time scan points projected onto the resistance–resilience plane, where resistance =* $1 - \frac{D}{D_{crit}}$ *and resilience = 1 − ρ* (contrast to van Meerbeek et al. (2021) resistance and resilience); green circles denote coexistence outcomes, red crosses denote regime shifts.*

Figure 4 quantifies regime-shift risk and highlights the main drivers of parametric instability—variability in phenology (the timing of seasonal biological events). The partial rank correlation coefficient (PRCC) shows how sensitive system stability at coexistence is to each parameter. Stability is measured using the dominant eigenvalue magnitude ρ* (indicating how quickly the system returns to equilibrium after disturbance), based on the Latin Hypercube sample (Step 10) in Figure 4A. Parameters with absolute PRCC above the threshold increase instability risk in ρ*. Only parasitoid attack rate β (the rate at which parasitoids attack hosts) and susceptible larval mortality μ_S (the death rate of larvae vulnerable to attack) surpass this threshold, but their effects differ; increasing β stabilises coexistence (positive PRCC, higher ρ*), while increasing μ_S lowers stability. Next most influential are parasitoid conversion efficiency η (the efficiency with which parasitoids develop within hosts) and adult parasitoid mortality μ_F (the death rate of adult parasitoids), indicating demographic sensitivity. Other parameters, which shift R_P in Figure 3, show moderate PRCC. R_P indicates potential for parasitoid invasion (whether a small number of parasitoids can successfully establish), while ρ* measures the speed of return to coexistence after disturbance.

Figure4B displays the regime-shift probability for each phenological scenario as well as for joint Latin Hypercube sampling. When all parameters are varied simultaneously, 81.5% of parameter sets force the coexistence equilibrium to shift. Season duration and beetle reproduction R_B are the most likely parameters to induce a regime shift when all parameters vary. Note that season duration variation alone did not show the shift in objective 4. However, in reality, season duration is linked to most phenologies; the figure 4B observation is most likely in the real world. Beetle overwinter survival σ_A and susceptible larval mortality μ_S also contribute significantly. In contrast, bird pressure B_t and foliage-memory φ do not induce regime shifts across sampling, similar to Figure 3, where, upon varying alone, B_t does not cross the $R_P = 1$ threshold from coexistence, and φ has minimal impact on R_P. The high joint shift probability, here all parameters vary jointly, the regime shift probability is relatively much higher than individual parameter values, which is an obvious assumption, but provides a comparison to individual parameter sensitivity.

Figure 4C projects all single-parameter scan points onto the resistance–resilience plane. Here, resistance is $1 - \frac{D}{D_{crit}}$ and resilience is $1 - \rho *$, which contrasts the resistance and resilience sensu van Meerbeek et al (2021) tested in later Figure 5. Points representing coexistence equilibrium cluster within a region of positive resistance and moderate resilience. Points for parasitoid-free or canopy-only regimes are more scattered and often show higher resilience. This occurs because the parasitoid-free equilibrium can have a smaller ρ* (faster return rate) than coexistence. Thus, resilient systems can remain in undesirable parasitoid-free regimes. This reflects the adaptive behaviour of this system's stability.

Together, these plots support hypothesis H2 that changes in certain phenologies, or parameter values, may serve as early warning signals of a shift to an undesirable equilibrium that still maintains stability, and identify season duration and beetle reproduction rate as one of the major drivers of the steady state. These results also confirm the importance of Van Meerbeek's definition of stability, that multiple stability metrics are needed to assess regime-shift risk. Based on these results, we provide Table 1 that is likely to guide early warning signal detection based on the phenology variability measurement in the field and its likely direction of future regime, followed by Figure 5 that shows how these parameter variability adapts resistance and resilience in the system.

Table 1: Candidate early-warning signal matrix for regime shift (Objective 2). Each phenological scenario is characterised by its regime-shift probability (one-at-a-time variation), PRCC with the dominant eigenvalue magnitude ρ, the destabilising direction of the corresponding parameter, the likely alternative equilibrium, and whether the phenological change is empirically observable. Parameters with regime-shift probability > 20% or |PRCC| > 0.25 that are empirically observable are classified as early-warning candidates*

Parameter	Potential early warning signals	Likely alternative equilibrium
T (Season length)	Longer season → lower R_P	Parasitoid-free
σ_A (Beetle overwinter)	Higher survival → higher beetle density	Parasitoid-free
σ_F (Parasitoid overwinter)	Higher survival → higher parasitoid	Coexistence maintained
μ_S (Larval mortality)	Lower μ_S → more larvae survive	Parasitoid-free
β (Parasitoid attack rate)	Higher β → stronger parasitism	Coexistence maintained
R_B (Beetle reproduction)	Higher R_B → stronger outbreaks	Parasitoid-free

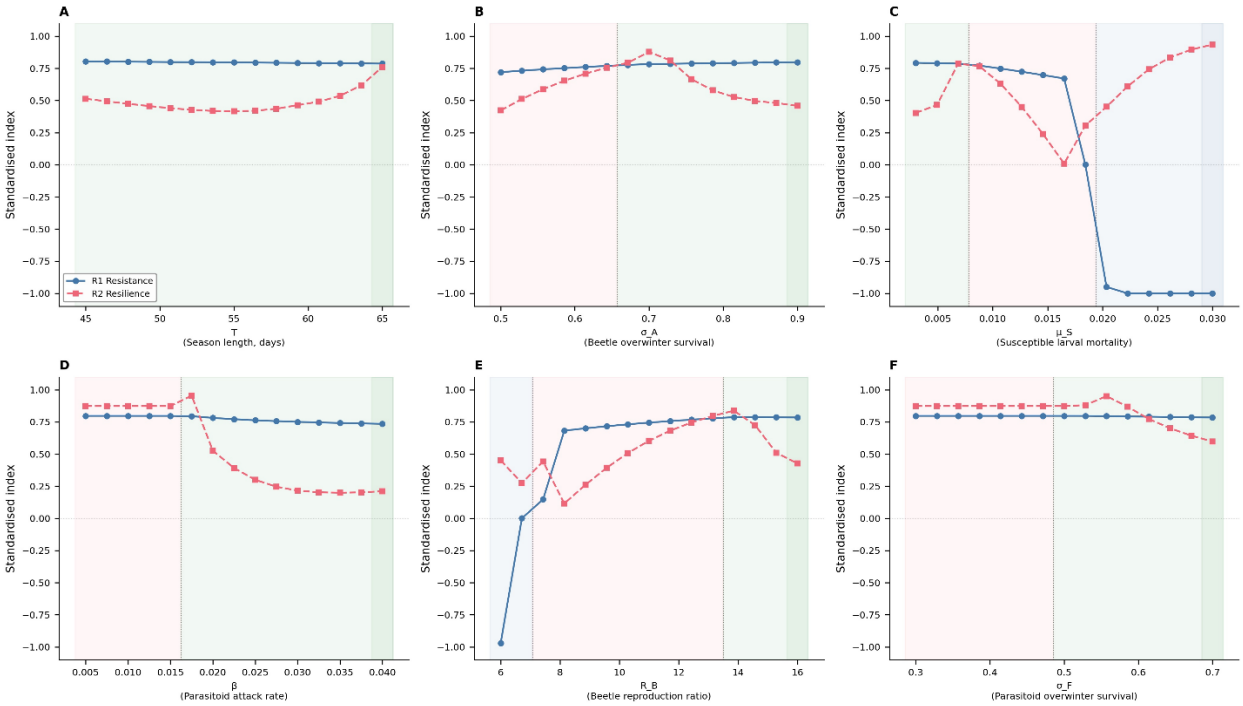

Figure 5: Resistance (R1) and resilience (R2) across the six candidate early-warning parameters identified in Figures 3-4 and Table 1. (A) Season duration T. (B) Beetle overwinter survival σ_A. (C) Susceptible larval mortality μ_S. (D) Parasitoid attack rate β. (E) Beetle reproduction ratio R_B. (F) Parasitoid overwinter survival σ_F. Background shading indicates the equilibrium regime (green: coexistence; pink: parasitoid-free; blue: canopy-only); dashed vertical lines mark transcritical bifurcation boundaries from Figure 2

Figure 5 helps identify the distinct dynamics of resistance (R1) and resilience (R2) sensu Van Meerbeek (2021) across the six parameters identified as potential early warning signal detectors through their covariance. The main purpose of this result is to validate our previous resistance and resilience metric, which helps us validate the previous set of results in light of adaptive stability theory. Also, it provides a comparative aspect to the result. Within the coexistence regime, R1 remains consistently high across all parameters. In contrast, R2 exhibits considerable variation within the same regime, indicating that the coexistence equilibrium can adapt to different resilience with the variability of these phenologies, while resistance can only change upon regime shift. Variability in susceptible beetle mortality is the only case where resilience R1 collapses. This is again another obvious assumption, as resistance is only observed when a perturbation is present; however, it still shows the differential behaviour between stability components in this system. Observation of this obvious trend also helps validate our result from our model.

3.3 Objective 3: Optimal Control Strategy Comparison

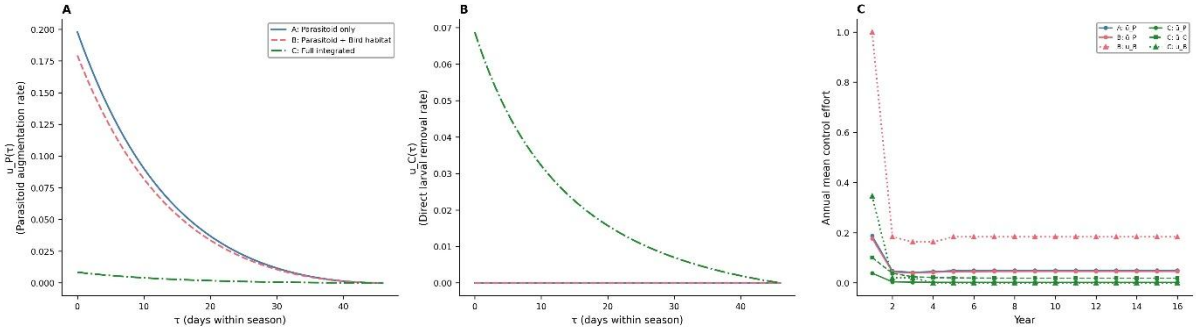

Figure 6: Optimal control allocation under three management scenarios. (A) Terminal within-season parasitoid augmentation rate $u_{P(\tau)}$. (B) Terminal within-season direct larval removal rate $u_{C(\tau)}$; active only in Scenario C. (C) Annual mean control effort over the 8-year planning horizon: mean within-season parasitoid augmentation $\bar{u}_P$ (circles, solid), mean within-season direct removal $\bar{u}_C$ (squares, dashed), and annual bird habitat enhancement u_B (triangles, dotted).

Figure 6 shows how the timing and type of management affect cost savings in this eco-epidemic system to meet objective 3. Panels A and B show changes in control actions within each year. Panel C compares average effort per year per strategy. Within each season, the three strategies differ in intervention focus and sequence. Strategy A relies solely on early parasitoid augmentation, i.e., the intentional release of parasitoid (u_P , Figure 6A) to maximise impact. Strategy B also begins with early parasitoid release, but later adds bird habitat enhancement, which reduces the need for additional augmentation. In contrast, Strategy C employs intensive early-season larval removal (Figure 6B), outperforming both parasitoid release and habitat improvements. Over multiple years (Figure 6C), effort allocation and cost trends diverge by strategy. All strategies have a peak within-season control in year 1, declining and stabilising as the managed equilibrium is approached. In Strategy B, bird habitat enhancement (u_B) is key: early, cost-effective suppression is enabled by initial investment, which transitions to low maintenance and reduces future costs. Strategy C instead uses moderate u_B in year 1, but emphasises larval removal and parasitoid augmentation for short-term control, resulting in lower long-term maintenance costs once stabilised. Strategy A, in contrast, remains reliant on parasitoid release without bird habitat investment or intensive larval removal, leading to different cost dynamics over time. Each strategy's approach to initial investment, maintenance, and intervention mix yields unique cost-saving patterns. This highlights how explicit choices in management strategy shape both short- and long-term cost efficiency.

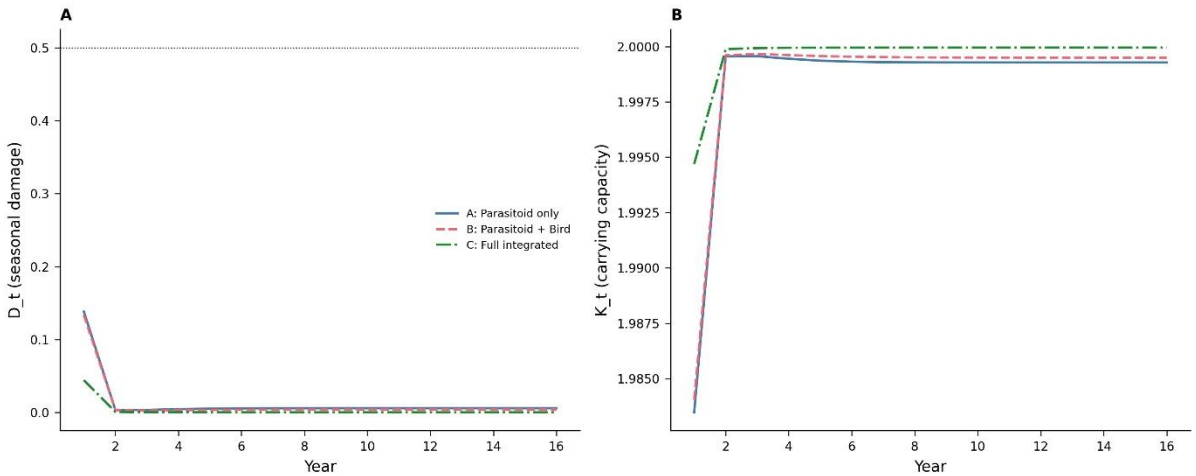

Figure 7: Multi-year state trajectories under three management scenarios. (A) Seasonal cumulative defoliation D_t and (B) carrying capacity K_t over the planning horizon. Dashed horizontal line in (A) marks the damage threshold $D_{crit} = 0.5$.

Figure 7 presents multi-year ecological outcomes for each management strategy, beginning from an outbreak state. Panel A shows seasonal cumulative defoliation (D_t), while Panel B shows carrying capacity (K_t). All strategies reduce defoliation below the threshold D_{crit} in the first year, but Strategy C converges fastest, Strategy B is intermediate, and Strategy A stabilises most slowly, with the most severe damage. Carrying capacity recovery follows the same ranking. Suppressing damage early reduces total defoliation, lessens the foliage-memory penalty (Equation 8), and lowers future intervention needs. For O3, the cost ranking from Figure 6 matches the ecological trajectory: the lowest-cost strategy achieves the fastest suppression and canopy recovery. According to the lexicographic decision rule outlined in the methodology, only Strategy C achieves local asymptotic stability at the managed equilibrium ($\rho^* = 0.98 < 1$). Strategies A ($\rho^* = 1.19$) and B ($\rho^* = 1.10$) exceed the stability boundary, rendering them infeasible despite reducing damage below D_{crit}. Strategy C is therefore the only feasible approach and also yields the minimum-cost outcome (J = 14.0, compared to 48.2 for A and 44.7 for B).

4. Discussion

Climate-driven phenological variability alters both persistence and control within multitrophic systems (Renner & Zohner, 2018). This study models the Black alder–leaf beetle–parasitoid–generalist bird ecoepidemic system. The aim of this modelling-based, theory-driven study was threefold. It sought to identify the most probable stable equilibria across all phenological scenarios, quantify the likelihood of the system shifting from its current coexistence equilibrium to an undesirable state, and determine the minimal effort required for three pest control strategies targeting the leaf beetle to prevent regime shifts across parameter space. Our methodology involves process-based modelling. It employs the system's structural properties, gathered from well-known phenologies, and uses optimal control theory to address these objectives. Results indicate that the system does not inherently favour the most desirable equilibrium of coexistence. We analysed nine phenological scenarios and found that desirable coexistence, characterised by parasitoid persistence, low defoliation, and a maintained canopy, occurs only within a limited region of parameter space representing phenological variability. The probability of regime shifts increases substantially when phenological parameters vary simultaneously. Only the fully integrated management strategy is found to prioritise feasibility across both seasons and years. The model and analyses are calibrated within possible parameter ranges derived from literature on these phenologies across habitats that house our focal system. Thus, these results do not constitute stand-level forecasts for any specific alluvial forests. Instead, they provide a general understanding of the system's stability properties, including resilience and resistance dynamics, from an eco-epidemic point of view. The stability properties resulting from our methodology may be observed in part or in full across European habitats containing the focal system.

The primary ecological implication of Objective 1 is that considering parasitoids as the main natural control for phylophagous leaf beetle populations may compromise the stability of coexistence under phenological variability. Within the parameter space defined by phenological variability, both parasitoid-free and canopy-only states are more frequently observed as stable equilibria than coexistence. Coexistence remains possible only if the parasitoid attack rate on beetles increases, both host and parasitoid populations exhibit sufficient overwintering survival, and generalist birds exert moderate predation pressure. These observations are consistent with existing host–parasitoid models, which indicate that coexistence within Nicholson–Bailey frameworks is limited to a narrow parameter range (Singh, 2023). This result also supports findings that phenological mismatch can disrupt tritrophic stability, even when pairwise interactions remain stable (Belitz et al., 2025). The greater likelihood of parasitoid-free and canopy-only stable equilibria in this system across phenological variability provides important insight for distinguishing between the desirability of coexistence in complex systems and the likelihood of convergence to a simpler, yet stable, structure. Ecological complexity is often linked to increased productivity due to higher species diversity, making it a desirable system attribute. In this focal system, parasitoids help reduce the risk of future beetle outbreaks, while generalist birds have access to other, more abundant prey. These findings often motivate forest habitat managers to introduce additional parasitoids to control outbreaks, as the attack rate per parasitoid cannot be directly managed.

The ecological theory of intraguild predation provides a detailed framework for validating such management decisions. This is especially useful when direct empirical evidence is lacking and can be done through analogies with the present findings. This theory suggests a higher-order predator suppresses an intermediate natural enemy while still reducing the shared prey (Rosenheim, 1998; Polis & Holt, 1992). In our focal system, the precise sign and magnitude of bird predation on parasitised Beetle larvae remain unmeasured. Parasitised beetle larvae continue feeding until death induced by the parasitoid (Stireman et al., 2006). They remain susceptible to bird predation throughout the delay interval. Within the model, bird predation reduces the total larval pool and consequently lowers defoliation. However, it also removes parasitised larvae before the koinobiont parasitoid completes development and emerges. Each parasitised larva removed in this manner constitutes a lost recruitment event for the endoparasitoid. If the release of the parasite is targeted at the foraging aggregation of the beetles, the attack rate might increase. This is a signal the parasitoid tracks to find host beetle larvae (Stireman et al., 2006). We advise the central management question to be increasing both the abundance and the efficiency of natural enemies. In this context, coexistence loss is more accurately described as a failure of parasitoid attack rather than merely insufficient parasite number.

Objective 2 extends Objective 1 by examining how phenological variation changes the probability of leaving coexistence. It quantifies the probabilities of each phenological variation that can cause a regime shift across any stable equilibrium. Therefore, while the prediction of Objective 1 is limited to the initial state where the parasitoid is present, Objective 2 covers cases where the parasitoid is initially absent. A key observational difference is that a single parameter can affect two equilibria through different mechanisms. For example, season length does not visibly affect coexistence equilibria within the range studied, except by drastically increasing defoliation damage. However, with a parasitoid-free initial state, season length can destabilise coexistence. This occurs because increased defoliation reduces next year's carrying capacity through a memory effect that reflects the accumulation of phytochemical defence that also changes the canopy as a feedback. Bird predation may reduce defoliation damage after a certain point. Its effect on coexistence is moderated by opposing intraguild predation effects if starting from the coexistence equilibrium. If starting from a parasitoid-free equilibrium, bird predation has no effect on parasitoid invasion risk. Phytochemical defence provides resistance and accumulates as plant memory, feeding back into the next year's carrying capacity for both beetle and parasitoid. However, it has little influence on the invasion or equilibrium class. This observation is subject to further study, as quantification challenges make the accumulation rate and impact on next year's growth poorly studied due to challenges in quantification. Since these results suggest that they are run on two different time scales, it establishes that resistance and resilience at each time scale shape stability differently in this system. Within-season trophic interactions determine parasitoid invasion risk. Year-to-year canopy defoliation memory of resistance determines the quality and persistence of long-term equilibrium. This supports a multi-metric stability framework (Van Meerbeek et al., 2021; Donohue et al., 2016). Invasion, resistance, and resilience capture different but complementary aspects of regime-shift risk. Consideration of examining resistance and resilience as two different stability metrics following the adaptive stability concept helped our understanding by examining the system's stability range via differential behaviour of these two metrics. High Resistance across much of the coexistence regime implies that outbreak damage can stay low, while resilience varies more strongly as recovery capacity can still erode under phenological change. The system can therefore look stable in the short term, yet become dynamically fragile in the long term, as per our findings. Together, all the results of objective 2 show that stability failure can occur through multiple distinct routes. Management should therefore not infer security from low damage alone, but evaluate invasion potential, recovery rate, and attractor-switch risk simultaneously.

In this study, we used Eigenvalue-based and trajectory-based metrics to represent mechanistically distinct aspects of stability. Within the Latin Hypercube sample, their correlation is positive but not perfect. Certain parameter sets demonstrate that the eigenvalue indicates rapid return, while trajectory-based resilience remains low due to finite perturbations nearing the basin boundary, and the reverse can also occur. From an ecological perspective, eigenvalue-based resilience characterises recovery from routine interannual fluctuations, whereas trajectory-based resilience characterises recovery from discrete, large-amplitude disturbances. This distinction similarly separates the equilibrium damage

margin from the transient resistance index. Reporting both metrics is essential for distinguishing chronic degradation risk from acute perturbation risk, which is directly relevant for selecting between sustained and emergency management strategies. In addition to the probability of regime shift under this adaptive stability, objective 2 aims to identify potential early warning signals using three joint criteria. First, the parameter must contribute substantially to regime-shift probability. Second, it must show a partial rank correlation with the eigenvalue that indicates a destabilising influence. Third, the associated phenological process must be empirically observable during a simulation experiment searching for system destabilisation. The candidate phenologies and their association with other phenologies as early warning signals, identified under Objective 2, are proposed as ranked monitoring priorities. They are not validated early warning signals capable of reliably indicating impending regime shifts, since our parameters come from multiple habitats. The three conditions set for early warning signal identification are not interchangeable during system analyses. For example, a parameter may drive eigenvalue ρ^* toward the stability boundary without greatly increasing regime-shift probability. Alternatively, a parameter may contribute to regime shifts but not be observable in practice. Among the identified candidates, season length is the most tractable. It can be measured with growing degree days or satellite-derived phenology. In contrast, directly estimating parasitoid attack rate or adult overwintering requires intensive field sampling. This has not yet been demonstrated at management-relevant scales in many systems (Nimalrathna et al., 2022). These variables provide a foundation for targeted monitoring designs and future studies on early warning signals of this system through phenologies.

The management implication of this study comes from the optimal theory-based result in Objective 3. The result suggests that it is not simply that one strategy reduced beetle damage faster than the others. Under the study's decision rule, a strategy is compared on cost only after it first proves ecologically feasible. In practical terms, this means the strategy must keep seasonal canopy loss below the critical damage level. It must also prevent the alder stand from slipping into a persistently degraded low-canopy condition and maintain a long-term managed state. In this state, parasitoids must remain present, and control must continue after small disturbances. All three strategies reduced canopy damage below the critical level within the first year. However, only the fully integrated strategy met those broader conditions. This strategy combined targeted larval removal, parasitoid release, and bird-habitat enhancement. The parasitoid-only strategy and the parasitoid-plus-bird strategy both improved short-term suppression but did not produce a self-maintaining low-damage state in the long run. This distinction separates temporary outbreak suppression from durable restoration. The key result is that the integrated strategy was not only the least costly, but was also the only one that reduced defoliation, supported canopy recovery, and preserved stable parasitoid-supported biological control. Cost ranking becomes meaningful only after the feasibility filter is passed. The ecological meanings used here follow your annex definitions of cumulative canopy damage, canopy availability, targeted larval removal, parasitoid release, and bird-habitat enhancement.

Our results from objective 3 also suggest that intervention timing matters for optimising the effort of the intervention. Early action prevents current-season beetle damage from being carried over into the next year as a structural canopy penalty. In this ecoepidemic system, reducing larval density early in the outbreak lowers the total defoliation accumulated over the season. This weakens the foliage-memory feedback. Heavy canopy loss otherwise reduces the forest's effective canopy support in the following year. As a result, the stand enters the next season in better condition. The amount of control needed later is reduced. All three management strategies, therefore, concentrate most of their effort at the beginning of the intervention period. Their efforts decline thereafter as the system moves toward a lower-damage managed state. Ecologically, this means the system responds best when control is applied before repeated feeding damage has accumulated enough to degrade future canopy recovery. The exact schedule may differ under alternative recovery assumptions, climate forcing, or cost structures. However, the general advantage of early, front-loaded action appears robust in this framework.

Our study integrates processes often studied separately, such as seasonal beetle feeding, delayed parasitoid development, bird predation on both healthy and parasitised larvae, interannual canopy damage, and diverse pest management. The main contribution is a comprehensive framework using a systems ecology view. This framework fills a literature gap on the system's structure-dependent multiple metrics that define adaptive stability properties. From a modelling perspective, our study

contributes by integrating an eco-epidemiological system that spans multiple trophic levels and shows intraguild predation. It also integrates optimal control theory through habitat management strategies as intervention measures. Most host-parasitoid models have focused just on insect roles and often exclude plant feedback or higher-order predators since the past century (Ji, Vasilyeva, Mbroh, & Sadovski, 2025; Nicholson & Bailey, 1935; Hassell et al., 1976). Many plant defence models look at how herbivory carries over across years (Hakes & Cronin, 2012), but they rarely assess management strategies. In addition to bridging these modelling gaps, our study also connects seasonal and annual phenology in a single model. This unification enables analysis of immediate damage control, long-term recovery, regime shifts, and explores the most effective management options under optimal control theory. However, as described in our methods, annex, and early warning signal discussion, varied sources for parameters limit our results to a general outline. This restricts us from implementing in a specific habitat directly without empirical data from that habitat. With long-term field data on beetle outbreaks, parasitoid numbers, bird predation, and other life-cycle stages from one habitat, our model could pinpoint regime shift chances, coexistence equilibrium, and disease-free equilibrium more precisely. Such data would also test the early warning signals and links we identified in our results. As mentioned earlier, our analysis is supported by systems structure using well-studied phenologies and their variability across space and time to map adaptive stability. Theories on adaptive stability and intraguild predation back up the findings (Van Meerbeek et al., 2021; Polis & Holt, 1992). Empirical, habitat-specific data for each habitat can determine (1) how habitat-specific parameter values narrow stability ranges for this system, and (2) help map the distribution of those habitats across the stability plane to inform further prioritisation of habitats at risk of regime shift. Therefore, this study aims to connect what we know about black alder phenology differences across habitats with future studies on spatial variability in adaptive stability of this system across black alder-rich habitats.

Author Contributions

SG: Conceptualisation, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualisation, Writing – original draft, Writing – review & editing. SS: Investigation, Validation, Writing – review & editing. All authors contributed to manuscript revision, read and approved the submitted version.

Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Data Availability Statement

No experimental datasets were generated or analysed in this study. All model parameters are derived from published sources cited in the text and listed in the Supplementary Material (Table S1). The numerical results presented are fully reproducible from the model equations, parameter values, and analysis procedures described in the article and Supplementary Material. The simulation code is available from the corresponding author upon reasonable request.

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Annex 1:

Table S 1: Model parameters and their rpresentative phenologies. A parameter that has anumber flagged with # shows a phenology that varies, and has been considered for parameter space definition during Latin hypercube search.

Parameter	Phenology Description	Value / Range	Reference
Module 1: Within-Season Eco-Epidemic Dynamics			
β	Parasitoid attack rate on susceptible larvae; Holling type II numerator. Phenology #8: variable parasitoid attack efficiency	0.001–0.005 ha fly^{-1} day^{-1} (calibrated baseline)	Calibrated to 1–5% parasitism prevalence; cross-system tachinid proxy (Hawkins, 1994; Stireman et al., 2006)
h	Parasitoid handling/incubation parameter; controls saturation of oviposition at high larval density	0–0.01 ha $larva^{-1}$ (baseline: 0.001)	Structural uncertainty parameter; scenario prior
c_B	Per-unit bird predation pressure consumption rate on larvae; Holling type II. Phenology #6: variable bird predation pressure	Tuned ha $pair^{-1}$ day^{-1}	Calibrated to match 3.1% daily predation rate on insect larvae (Remmel et al., 2011)
a_B	Bird half-saturation parameter: controls how predation saturates with total larval density (S+I)	0–0.01 ha $larva^{-1}$ (baseline: 0.0001)	Structural uncertainty parameter; scenario prior
μ_S	Background mortality of susceptible larvae from causes other than parasitism and bird predation. Phenology #5: decreased susceptible larval mortality	0.01–0.10 day^{-1} (baseline: 0.04)	General chrysomelid larval mortality ranges (Cornell & Hawkins, 1995); fungal mortality included
$u_C(\tau)$	Time-dependent targeted removal of larvae; management control variable	0–u_{C,max} day^{-1}	Management control variable
μ_I	Background/base line mortality of parasitised larvae; assumed slightly > μ_S due to parasitoid physiological burden	0.02–0.12 day^{-1} (baseline: 0.05)	Scenario prior; assumed > μ_S
δ	Parasitoid emergence rate; inverse of mean development time from	0.04–0.14 day^{-1} (baseline: 0.10)	Tachinid development 7–25 days on hosts

	infection to host death. Phenology #4: accelerated parasitoid development		(Hawkins, 1994; Stireman et al., 2006)
η	Parasitoids produced per killed host; solitary endoparasitoid biology of M. mutabilis	0.4–1.0 dimensionless (baseline: 1)	Solitary biology: 1 parasitoid per host; range reflects developmental losses (Stireman et al., 2006)
μ_F	Adult parasitoid fly natural mortality rate	0.05–0.15 day^{-1} (baseline: 0.10)	General tachinid adult longevity; exact value for M. mutabilis not found; scenario prior
$u_P(\tau)$	Time-dependent parasitoid augmentation rate; management control variable	0–u_{P,max} flies ha^{-1} day^{-1}	Management control variable
κ	Converts larval density to fractional defoliation rate; represents larval foraging rate on alder leaves	Scaled $(individuals \cdot day)^{-1}$	Calibrated: D_t ≈ 0.5 at typical outbreak densities (Dolch & Tscharntke, 2000)
T	Duration of larval vulnerability window. Phenology #1: extended/shortened season length under warming climate	45–65 days (baseline: 55)	Univoltine phenology of Beetle
Module 2: Annual Recurrence Map (Equations 5–8)			
R_B	Annual basic reproduction rate/ratio of the beetle; net fecundity after egg-to-adult survival. Phenology #9: variable beetle fecundity	1.5–20 dimensionless (baseline: 14)	Clutch size 20–70 eggs per female (Sivrikaya et al., 2020); net rate depends on overwinter survival
σ_A	Larva-to-adult overwintering survival of the beetle. Phenology #2: reduced winter severity (beetle)	0.3–0.8 dimensionless (baseline: 0.5)	Adults burrow into soil litter; exact survival not quantified for Beetle; scenario prior (Sivrikaya et al., 2020)
σ_F	Parasitoid overwinter survival (tachinid puparia). Phenology #3: reduced winter severity (parasitoid)	0.1–0.4 dimensionless (baseline: 0.2)	General tachinid puparia survival; exact value for M. mutabilis not found; scenario prior
B_t^{index}	Baseline generalist passerine bird pressure index. Phenology #6: variable bird predation pressure	0.5–2.0 pairs ha^{-1} (baseline: 1.0)	PECBMS annual forest bird indices; great tit ~1.1–1.4 pairs ha^{-1} in Central European floodplain (Gregory et al., 2007)
ξ	Scales bird-habitat management effort to its effect on local bird density	Scenario-dependent dimensionless	Nest-box provisioning effect (Mols & Visser,

			2002); exact value for FFH 91E0 not quantified
$\mathbf{u_B}(t)$	Annual bird-habitat enhancement; management control variable	0–u_{B,max} dimensionless	Management control variable
$\mathbf{K_0}$	Undamaged baseline carrying capacity of beetle (fully recovered canopy)	Scenario-dependent scaled larvae ha^{-1}	ICP Forests Level I network (Ferretti et al., 2021)
$\boldsymbol{\varphi}$	Damage-to-capacity penalty; accumulated phytochemical defence feedback. Phenology #7: accumulated phytochemical defence	0.001–0.05 dimensionless (baseline: 0.01)	0.9% growth loss per 1% defoliation (Ferretti et al., 2021); induced resistance ~133 days (Dolch & Tscharntke, 2000)
Objective Function and Control Bounds (Equation 9)			
$\boldsymbol{\omega_D}$	Weight on within-season running defoliation damage D(τ)	Dimensionless	Management preference
$\boldsymbol{\omega_S}$	Weight on susceptible larval density S(τ)	Dimensionless	Management preference
$\boldsymbol{\omega_T}$	Weight on terminal (end-of-season) damage D_t	Dimensionless	Management preference
$\mathbf{c_P}$	Unit cost of parasitoid augmentation	Dimensionless	Fraction of forest management budget
$\mathbf{c_C}$	Unit cost of targeted larval removal	Dimensionless	Fraction of forest management budget
$\mathbf{c_{B,cost}}$	Unit cost of bird-habitat conservation. Note: distinct from c_B (bird foraging rate in ODEs)	Dimensionless	Fraction of forest management budget
Stability Analysis Thresholds			
$\mathbf{D_{crit}}$	Functional damage threshold for canopy viability	0.50 dimensionless	50% seasonal defoliation threshold (Ferretti et al., 2021)
$\mathbf{K_{min}}$	Minimum acceptable carrying capacity for persistent canopy	0.5 K_0 scaled	Management threshold for persistent canopy degradation

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Annex 2: Stability Analyses

This annex presents the complete mathematical framework for the stability analysis of the hybrid seasonal–annual eco-epidemic model described in the main text (Equations 1-8). The long-term stability of this system is determined by the annual recurrence map rather than by the within-season ordinary differential equation (ODE) considered in isolation. The within-season ODE (Equations 1-4) characterises loss and accumulation processes over a finite interval $\tau \in [0, T]$. Setting $dS/d\tau = dI/d\tau = dF/d\tau = 0$ produces within-season pseudo-equilibria, which do not represent ecologically relevant steady states, since the system is reset annually through recruitment, overwinter survival, and foliage recovery equations (Equations 5-8). Multi-year dynamic-including outbreak recurrence, parasitoid persistence, and tipping risk,are governed by the annual map.

The stability analysis is analytically well-posed but does not admit a closed-form solution. Specifically, the annual map G and its structure, existence conditions for fixed points, the Jacobian J_G of the annual map (derived via the chain rule and variational equations), the parasitoid invasion threshold R_P as an eigenvalue of J_G, local stability criteria (spectral radius less than one), and bifurcation conditions (eigenvalue crossing the unit circle) are all derived analytically. In contrast, the seasonal flow Φ_T and its derivatives are evaluated numerically, as the Holling Type II functional responses render the within-season ODE nonlinear and analytically intractable. This approach, combining analytic stability conditions with numerical evaluation of the seasonal flow, is standard and mathematically rigorous for hybrid eco-epidemic models (Kot, 2001; Kuznetsov, 2004).The analysis proceeds through a ten-step framework, each anchored to the specific theorem, criterion, or mathematical principle that it employs.

Step 1: Definition of the annual state and the seasonal flow operator:

The annual state vector is defined as $x_t = (A_t, F_t, K_t) \in \mathbb{R}^3_+$, representing, respectively, the recruiting beetle cohort (individuals ha^{-1}), the overwintering parasitoid population (individuals ha^{-1}), and the implicit alder carrying capacity (individuals ha^{-1}) at the start of year t.

The within-season ordinary differential equation system (Equations 1-4 of the main text) defines a continuous-time dynamical system over $\tau \in [0, T]$, where T is the duration of the larval vulnerability window. This system operates on the within-season state vector $y(\tau) = (S(\tau), I(\tau), F(\tau), D(\tau))^T$, where S is susceptible larval density, I is parasitised larval density, F is adult parasitoid density, and D is cumulative defoliation damage.

The seasonal flow operator Φ_T maps initial conditions and parameters to terminal states:

$$(S(T), I(T), F(T), D(T)) = \Phi_t(S(0), I(0), F(0), D(0);\ B_t, \theta, u(\cdot)) \tag{12}$$

Here, θ represents the vector of biological parameters and $u(\cdot) = (u_P(\tau), u_C(\tau))$ specifies the within-season control profiles. The initial conditions are determined by the annual state via Equation (5):

$$S(0) = R_B\, A_t / (1 + A_t/K_t),\ \ I(0) = 0,\ \ F(0) = F_t,\ \ D(0) = 0$$

The forward-invariant domain on which Φ_T operates is:

$$\Omega = \{(S, I, F, D) \in \mathbb{R}^4 : S \geq 0, I \geq 0, F \geq 0, D \geq 0\}$$

Theorem used: Picard–Lindelöf (Cauchy–Lipschitz) existence and uniqueness theorem.

Statement (Hartman, 1964, Theorem II.1.1; Coddington and Levinson, 1955, Theorem 2.2): If f(y) is locally Lipschitz continuous in y on an open set $U \subseteq \mathbb{R}^n$, then for each $y_0 \in U$ the initial value problem $y' = f(y)$, $y(0) = y_0$ has a unique local solution.

Application.

The right-hand side $f(y; \theta, B_t)$ of Equations (1)–(4) contains two rational terms with denominators $(1 + hS)$ and $(1 + a_B(S + I))$. For $S \geq 0$ and $I \geq 0$, both denominators are strictly positive (bounded below by 1), so the rational terms are smooth (in fact $C\infty$) functions of the state variables on Ω. All remaining terms are polynomial in the state variables. Therefore, f is locally Lipschitz on any bounded subset of Ω. Since the seasonal interval [0, T] is finite and the state variables remain bounded on this interval (the ODE is dissipative: $dS/d\tau \leq 0$, $dI/d\tau$ is bounded by inflow from parasitism, and D grows at most linearly), the Picard–Lindelöf theorem guarantees that a unique solution $y(\tau)$ exists for all $\tau \in [0, T]$ for each initial condition in Ω.

Positivity invariance.

Positivity invariance of Ω is established by verifying that each compartment boundary is non-repelling under the vector field (Nagumo, 1942; Blanchini, 1999). Specifically:

At S = 0: $dS/d\tau = 0$ (all terms in Equation 1 contain S as a factor), so the S = 0 face is invariant.

At $I = 0$: $\frac{dI}{d\tau} = \frac{\beta FS}{1 + hS} \geq 0$ for $S \geq 0$, $F \geq 0$, so trajectories cannot exit through the I = 0 face into I < 0.

At $F = 0$: $\frac{dF}{d\tau} = \eta\delta I + uP(\tau) \geq 0$ for $I \geq 0$ and $u_P \geq 0$, so trajectories cannot exit through the F = 0 face.

At $D = 0$: $\frac{dD}{d\tau} = \kappa(S + I) \geq 0$ for $S \geq 0$, $I \geq 0$, so D remains non-negative.

Therefore, Ω is forward-invariant under the flow, and Φ_T is well-defined on Ω for each year given B_t and controls.

Step 2: Construction of the annual recurrence map

From Equations (5)–(8) of the main text and the seasonal flow operator Φ_T defined in Step 1, the annual map $G: \mathbb{R}^3_+ \rightarrow \mathbb{R}^3_+$ is constructed as:

$$x_{t+1} = G(x_t) = (\sigma_a\, S(T;\ x_t),\ \sigma_F\, F(T;\ x_t),\ K_0\, exp(-\phi\, D(T;\ x_t))) \tag{13}$$

where $S(T;\ x_t), F(T;\ x_t)$, and $D(T;\ x_t)$ denote the terminal values S(T), F(T), D(T) obtained by evaluating Φ_T at the initial conditions $S(0) = R_B A_t/(1 + A_t/K_t), I(0) = 0, F(0) = Ft, D(0) = 0$.

Mathematical basis.

This construction is the Poincaré (stroboscopic) return map of the periodically forced system (Wiggins, 2003, §1.2; Kot, 2001, Chapter 13). The Poincaré map converts a continuous-time system with periodic

forcing into a discrete-time dynamical system by sampling the state at the end of each period. For seasonally structured ecological models, this approach is standard (Elaydi, 2005; Kuznetsov, 2004) and reduces the infinite-dimensional problem of analysing continuous trajectories to the finite-dimensional problem of analysing the iteration $\boldsymbol{x_{t+1}} = G(\boldsymbol{x_t})$.

Step 3: Identification of fixed points (annual equilibria) of the map

Fixed points $x^* = G(x^*)$are obtained by solving the nonlinear system $H(x) = G(x) - x = 0$ numerically via Newton iteration or continuation methods (Allgower and Georg, 2003). The following ecologically distinct fixed points are sought.

(i) Trivial (extinction) fixed point: Trivial (extinction) fixed point $x_0^* = (0, 0, K_0)$: $A^* = 0$, $F^* = 0$, $K^* = K_0$ This corresponds to the beetle-free state. When A = 0, no larvae are present, so $S(T) = 0,\backslash quad\ F(T) = F_0, e^{\{-\backslash mu_F T\}}$(parasitoid decays without hosts), and D(T) = 0. Hence $A_{t+1} = 0,\quad F_{t+1} = \sigma_F\, F_0\, e^{-\mu_F T} \to 0$ as $F_0 \to 0,$ and $K_{t+1} = K_0 \exp(0) = K_0$..This fixed point is relevant only as a boundary reference.

(ii) Parasitoid-free fixed point $x_1^* = (A^*, 0, K^*)$: The beetle persists at a density regulated by bird predation, baseline mortality, and foliage feedback, but no parasitoid is present ($F^* = 0$). When F = 0, the parasitism term $\beta FS/(1 + hS)$ vanishes, so $I(\tau) \equiv 0$ throughout the season. The susceptible larvae decay under natural mortality and bird predation only, and $A_{t+1} = \sigma_A\ S(T)$. The self-consistency condition $A^* = \sigma_A\ S(T; A^*, 0, K^*)$ uniquely determines A^* for given K^* and B_t because the Beverton–Holt function $S(0) = R_B A/(1 + A/K)$ is monotonically increasing and concave in A (Beverton and Holt, 1957), and the subsequent within-season decay is monotonically decreasing. This is the state from which parasitoid invasion is assessed.

(iii) Coexistence (endemic parasitism) fixed point $x_2^* = (A^*, F^*, K^*)$ with $F^* > 0$: Both the beetle and the parasitoid persist, and the parasitoid exerts ongoing biological control. This is the management target. At this fixed point, the parasitoid population is sustained by sufficient susceptible host density to overcome within-season mortality and overwinter attrition.

Existence conditions for each fixed point are derived implicitly,, the terminal states S_T, F_T, D_T depend on x^* through Φ_T, so closed-form expressions are generally unavailable due to the nonlinearity of the Holling Type II terms. The existence and number of fixed points are determined by Newton iteration with warm-start initialisation (iterating the map forward ~100–200 years to approach the attractor, then refining with Newton's method), and their dependence on parameters can be tracked via numerical continuation (Allgower and Georg, 2003).

Step 4: Local stability of annual fixed points via the Jacobian of the annual map.

Criterion used: Spectral radius criterion for discrete-time maps.

Theorem (Elaydi, 2005, Theorem 4.15): A fixed point x^* of a discrete-time map $x_{t+1} = G(x_t)$ is locally asymptotically stable if and only if all eigenvalues of the Jacobian matrix $J_G(x^*) = \partial G/\partial x|_{x^*}$ lie strictly inside the unit circle in the complex plane, i.e., $\max_i |\lambda_i| < 1$, where λ_i are the eigenvalues of $J_G(x^*)$.

Justification: Discrete-time Hartman–Grobman linearisation theorem.

Theorem (Kuznetsov, 2004, Chapter 2; Palis and de Melo, 1982): Near a hyperbolic fixed point (no eigenvalues on the unit circle), the nonlinear map G and its linearisation $x \to J_G(x^*)(x - x^*)$ are topologically conjugate, i.e., there exists a homeomorphism that maps orbits of one to orbits of the other, preserving orientation. This guarantees that the local qualitative dynamics (stability, instability, saddle structure) of the nonlinear map are completely determined by the eigenvalues of $J_G(x^*)$.

Important distinction: This is the correct stability test for the annual recurrence. The Routh–Hurwitz criterion ($\mathrm{Re}(\lambda_i) < 0$) applies to continuous-time systems and would be appropriate only for the within-

season ODE in isolation. For the discrete-time annual map, the unit circle (not the imaginary axis) is the stability boundary.

Definition of the Jacobian.

The Jacobian of the annual map at a fixed-point x* is the 3 × 3 matrix:

$$J_x(x^*) = \partial G/\partial x\,|_{x^*} \tag{14}$$

By the chain rule, the entries of J_G decompose into two components: (a) analytic derivatives of the initial conditions with respect to the annual state, and (b) numerical derivatives of the seasonal terminal values with respect to the initial conditions.

Component (a): Initial-condition derivatives from Beverton–Holt recruitment.

From $S(0) = R_B A_t/(1 + A_t/K_t)$, direct differentiation yields (Beverton and Holt, 1957): $\frac{\partial S(0)}{\partial A_t} = \frac{R_B}{(1+\backslash dfrac A_t K_t)^2}$ (15a); $\frac{\partial S(0)}{\partial K_t} = \frac{R_B\, A_t^2}{K_t^2\left(1+\frac{A_t}{K_t}\right)^2}$ (15b) $\frac{\partial S(0)}{\partial F}_t = 0$ (15c)

Derivation of Equation (15a): Write $S(0) = R_B\, A\, K / (K + A)$ where the subscripts t are suppressed for clarity. Then $\partial S(0)/\partial A = R_B\, K / (K + A) - R_B\, A\, K / (K + A)^2 = R_B\, K^2 / (K + A)^2 = R_B / (1 + A/K)^2$.

Derivation of Equation (15b): $\partial S(0)/\partial K = R_B\, A\, (K + A) - R_B\, A\, K) / (K + A)^2 = R_B\, A^2 / (K + A)^2 = R_B\, A^2 / [K^2(1 + A/K)^2]$.

Additionally, from Equation (5): $\partial F(0)/\partial F_t = 1$, and $\partial F(0)/\partial A_t = \partial F(0)/\partial K_t = 0$, and $\partial I(0)/\partial x_t = 0$, $\partial D(0)/\partial x_t = 0$ (since I(0) = 0 and D(0) = 0 are fixed constants each year).

Component (b): Flow sensitivities.

The derivatives $\partial S_T/\partial S(0)$, $\partial S_T/\partial F(0)$, $\partial F_T/\partial S(0)$, $\partial F_T/\partial F(0)$, $\partial D_T/\partial S(0)$, $\partial D_T/\partial F(0)$ are obtained from the variational equations (Step 5). Note: only the sensitivities with respect to S(0) and F(0) are required, because I(0) = 0 and D(0) = 0 are not functions of the annual state x_t.

Assembly of the full Jacobian.

Combining components (a) and (b) via the chain rule, the three rows of J_G are:

$$\partial A_{t+1}/\partial x_t = \sigma_a\,(\partial S_t/\partial S(0) \cdot \partial S(0)/\partial x_t + \partial S_t/\partial F(0) \cdot \partial F(0)/\partial x_t) \tag{16a}$$

$$\partial F_{t+1}/\partial x_t = \sigma_F\,(\partial F_t/\partial S(0) \cdot \partial S(0)/\partial x_t + \partial F_t/\partial F(0) \cdot \partial F(0)/\partial x_t) \tag{16b}$$

$$\partial K_{t+1}/\partial x_t = -\phi\, K_{t+1}\,(\partial D_t/\partial S(0) \cdot \partial S(0)/\partial x_t + \partial D_t/\partial F(0) \cdot \partial F(0)/\partial x_t) \tag{16c}$$

Derivation of Equation (16c): From $K_{t+1} = K_0 \exp(-\phi\, D_T)$, differentiate: $\partial K_{t+1}/\partial x_t = K_0 \exp(-\phi\, D_T) \cdot (-\phi) \cdot \partial D_T/\partial x_t = -\phi\, K_{t+1} \cdot \partial D_T/\partial x_t$. The chain rule then expands $\partial D_T/\partial x_t$ through the initial conditions.

The Jacobian is thus analytically defined but numerically evaluated—this is standard and mathematically rigorous for hybrid eco-epidemic models with nonlinear functional responses (Kot, 2001).

Step 5: Derivatives of the seasonal flow via variational (sensitivity) equations

Purpose.

Compute the partial derivatives of the seasonal terminal values (S_T, I_T, F_T, D_T) with respect to the initial conditions (S(0), I(0), F(0), D(0)), which are required for assembling J_G in Step 4.

Mathematical framework: Tangent linear model (variational equations).

Principle (Hartman, 1964, Chapter V; Khalil, 2002, §3.1): Let $y(\tau)$ be a solution of $y' = f(y)$ with $y(0) = y_0$. If f is continuously differentiable, then the sensitivity of $y(\tau)$ to perturbations in y_0 is captured by

the sensitivity (monodromy) matrix $M(\tau) = \partial y(\tau)/\partial y(0)$, which satisfies the 4×4 linear matrix ODE: $M'(\tau) = \left[\frac{\partial f}{\partial y}\right]_{y(\tau)} \cdot M(\tau), \quad M(0) = I_4$ (17)

where $\partial f/\partial y$ is the Jacobian of the within-season ODE right-hand side evaluated along the trajectory $y(\tau)$, and I_4 is the 4×4 identity matrix. Equation (17) is a linear, non-autonomous ODE (the coefficient matrix varies with τ through the reference trajectory $y(\tau)$), integrated numerically alongside the state equations from $\tau = 0$ to $\tau = T$.

Extraction of required sensitivities.

The terminal value $M(T)$ provides all required sensitivities. Only the columns corresponding to initial conditions that depend on the annual state are needed. Since $I(0) = 0$ and $D(0) = 0$ are fixed constants (not functions of x_t), only columns 1 (derivatives w.r.t. $S(0)$) and 3 (derivatives w.r.t. $F(0)$) of $M(T)$ are required:

$$\begin{aligned} &\partial S_t/\partial S(0) = M_{1,1}(T), \quad \partial S_t/\partial F(0) = M_{1,3}(T) \\ &\partial F_t/\partial S(0) = M_{3,1}(T), \quad \partial F_t/\partial F(0) = M_{3,3}(T) \\ &\partial D_t/\partial S(0) = M_{4,1}(T), \quad \partial D_t/\partial F(0) = M_{4,3}(T) \end{aligned} \tag{18}$$

These derivatives, combined with the analytic initial-condition derivatives (Equations 15a–c), fully determine the Jacobian $J_G(x^*)$ via Equations (16a–c).

Extension to parameter sensitivities.

The same variational framework extends to parameter sensitivities $\partial y(T)/\partial\theta$ by augmenting Equation (17) with the inhomogeneous term $\partial f/\partial\theta$:

$$d/d\tau\, [\partial y/\partial\theta] = [\partial f/\partial y] \cdot [\partial y/\partial\theta] + \partial f/\partial\theta, \quad \partial y(0)/\partial\theta = 0$$

This augmented system supports the global sensitivity analysis in Step 10.

Step 6: Parasitoid invasion threshold from the annual map

Definition.

The parasitoid invasion threshold R_P (Equation 11 of the main text) is defined as the linearised one-year growth factor of the parasitoid at the parasitoid-free fixed point $x_1^* = (A^*, 0, K^*)$:

$$R_P \equiv \left(\frac{\partial F_{\{t+1\}}}{\partial F_t}\right)\Big|\{\boldsymbol{x}^{1*}\} = \sigma_F \cdot \left(\frac{\partial F(T)}{\partial F(0)}\right)\Big|\{S(0) = \hat{S}, I(0) = 0, F(0) = 0\} \tag{11}$$

where $\partial F(T)/\partial F(0)$ is the variational sensitivity obtained from the sensitivity matrix $M(T)$ (Step 5), evaluated along the parasitoid-free seasonal trajectory.

Mathematical basis: Block-triangular structure of J_x at x_1^*.

Proof that R_P is an eigenvalue of $J_x(x_1^*)$. At the parasitoid-free fixed-point $x_1^* = (A^*, 0, K^*)$, the parasitoid is absent ($F^* = 0$) and no larvae are parasitised ($I^* = 0$ throughout the season). Under the parasitoid-free seasonal trajectory, $F(\tau) \equiv 0$ and $I(\tau) \equiv 0$ for all $\tau \in [0, T]$. Examining the within-season Jacobian $\partial f/\partial y$ along this trajectory:

The entry $\partial(dI/d\tau)/\partial S = \beta F/(1 + hS)^2 = 0$ (because $F = 0$), and the entry $\partial(dF/d\tau)/\partial S = 0$ (Equation 3 does not depend on S directly). Consequently, the variational equation for $M_{3,1}(\tau) = \partial F(\tau)/\partial S(0)$ receives no forcing from the S-perturbation ($M_{3,1}(0) = 0$ and the coupling coefficients are zero), so $M3,1(T) = \partial FT/\partial S(0) = 0$ at x_1^*.

In the annual state ordering $x = (A, F, K)$, the second row of J_G at x_1^* therefore becomes:

$$\partial F_{t+1}/\partial A_t = \sigma_F \cdot \partial F_t/\partial S(0) \cdot \partial S(0)/\partial A_t = \sigma_e \cdot 0 \cdot (\cdot) = 0$$
$$\partial F_{t+1}/\partial F_t = \sigma_F \cdot \partial F_t/\partial F(0) = R_P$$

$$\partial F_{t+1}/\partial K_t = \sigma_F \cdot \partial F_t/\partial S(0) \cdot \partial S(0)/\partial K_t = 0$$

The second row of $J_G(x_1^*)$ is [0, R_P, 0]. The characteristic polynomial of the 3 × 3 Jacobian therefore factors as:

$$\det(J_x - \lambda l) = (R_P - \lambda) \cdot \det([J_{aa} - \lambda, J_{ak}; J_{ka}, J_{kk} - \lambda])$$

where J_{AA}, J_{AK}, J_{KA}, J_{KK} are the entries of the (A, K) beetle–foliage subsystem block. Therefore, R_P is always an eigenvalue of J_x at the parasitoid-free fixed point—this is an exact structural result, not an approximation. The remaining two eigenvalues govern the stability of the resident beetle–foliage dynamics.

Ecological interpretation.

R_P measures the expected number of new parasitoid adults produced per parasitoid introduced over one full annual cycle (within-season parasitism, host mortality, parasitoid emergence, and overwinter survival) at the resident beetle density. This interpretation connects the next-generation logic of van den Driessche and Watmough (2002) to the Floquet theory for periodically forced systems (Wiggins, 2003, §1.2).

Stability implications.

The parasitoid-free fixed point x_1^* is locally asymptotically stable as an annual equilibrium if and only if (a) $R_P < 1$, and (b) the beetle–foliage eigenvalues of the resident subsystem also lie inside the unit circle. At $R_P = 1$, a transcritical bifurcation occurs: the coexistence fixed point x_2^* becomes biologically feasible ($F^* > 0$) and, generically, inherits stability for $R_P > 1$ while x_1^* loses stability in the parasitoid direction.

Approximate formula.

When the within-season dynamics are evaluated under the simplifying assumption that $S \approx \acute{S}$ (constant host density), with constant bird pressure B_t and including the Holling Type II saturation correction, the rigorous definition reduces to the interpretable next-generation approximation: $R_P \approx$

$$\frac{\eta\,\delta\,\beta\,\hat{S}}{(1+h\hat{S})\cdot\mu_F\cdot\left(\mu_I+\delta+u_C+\frac{c_B B_t}{1+a_B\hat{S}}\right)} \quad (11a)$$

Note the factor $(1 + h\acute{S})$ in the denominator, which arises from the Holling Type II functional response. This factor is negligible when $h\acute{S} \ll 1$ but should be retained when handling time h is appreciable relative to the inverse of the host density.

Step 7: Identification of tipping points as bifurcation boundaries of the annual map

The operational definitions of resistance, resilience, adaptive stability, tipping, and the strategy comparison protocol are given in Section 2.2 of the main text; Annex 2 provides the mathematical derivations underlying those quantities.

Criterion used: Bifurcation theory for discrete-time maps.

Theorem (Kuznetsov, 2004, Chapter 4): A fixed point $x^*(p)$ of a parametrised discrete-time map $x_{t+1} = G(x_t; p)$ undergoes a bifurcation at $p = p_c$ when an eigenvalue (or a pair of eigenvalues) of $J_G(x^*(p))$ crosses the unit circle. The generic codimension-one bifurcations of discrete-time maps are: (a) saddle-node (fold), (b) period-doubling (flip), and (c) Neimark–Sacker (discrete Hopf).

Application.

A tipping parameter p (chosen from $\{R_B, \phi, c_B, \beta, u_{P,max}, u_{B,max}\}$ according to the management question) is varied, and the eigenvalues $\lambda_i(p)$ of $J_G(x^*(p))$ are tracked. The regulated (low-damage) fixed point $x^*(p)$ is locally asymptotically stable as long as $\max_i |\lambda_i(p)| < 1$. A tipping boundary occurs at the critical value $p = p_c$ where:

$$\max_i |\lambda_i(pc)| = 1 \quad (19)$$

Classification of tipping types.

(a) Saddle-node (fold) tipping: A real eigenvalue crosses +1. This corresponds to the collision and disappearance of the low-damage and high-damage fixed points. Ecologically, the stable low-damage equilibrium ceases to exist and the system collapses abruptly to a high-damage state.

(b) Flip (period-doubling) tipping: A real eigenvalue crosses −1. This corresponds to the onset of biennial outbreak cycling. Ecologically, this produces alternating outbreak and recovery years—the system oscillates between high-damage and low-damage states with period 2.

(c) Neimark–Sacker (discrete Hopf) tipping: A complex conjugate pair of eigenvalues crosses the unit circle. This corresponds to the onset of quasi-periodic outbreak oscillations. Ecologically, this produces irregular multi-year outbreak cycles whose period is generally incommensurate with the annual timescale.

Functional tipping criteria.

In addition to the dynamical tipping boundary (Equation 19), a functional tipping criterion is defined to capture ecological collapse that may not coincide exactly with the mathematical bifurcation. Two functional thresholds are monitored: (i) the carrying capacity K_t falls below a regeneration threshold K_{min} for $m \geq 3$ consecutive years, indicating persistent canopy degradation; or (ii) the long-run mean damage exceeds a critical threshold D_{crit}. The functional tipping boundary is located relative to the dynamical bifurcation boundary to assess whether ecological collapse precedes or follows the loss of mathematical stability.

Multistability and basin boundaries.

For parameter values near p_c, the annual map is iterated forward from multiple initial conditions to detect multistability—the coexistence of a low-damage attractor and a high-damage attractor (or stable cycle) separated by a basin boundary. If multistability exists, tipping can occur by perturbation (basin crossing) even before the dynamical bifurcation, and the relative basin sizes are reported as a measure of tipping risk.

Step 8: Optimal control via Pontryagin's Maximum Principle and the forward–backward sweep method

Purpose.

Determine the optimal allocation of within-season control effort (parasitoid augmentation u_P and targeted larval removal u_C) that minimises the combined ecological and economic cost.

Theorem used: Pontryagin's Maximum Principle (PMP).

Theorem (Pontryagin et al., 1962; Fleming and Rishel, 1975): For a control system $y' = f(y, u)$ with cost functional $J = \int L(y, u)\, d\tau + \Psi(y(T))$, there exist costate variables $\lambda(\tau)$ such that, at the optimum, the Hamiltonian $H = L + \lambda^T f$ satisfies: (a) the state equations $y' = \partial H/\partial\lambda = f$; (b) the costate equations $\lambda' = -\partial H/\partial y$; (c) the stationarity condition $\partial H/\partial u = 0$ at the optimal control; and (d) the transversality condition $\lambda(T) = \partial\Psi/\partial y(T)$.

Sign convention (minimisation).

Convention adopted: The Hamiltonian is defined in the standard minimisation form $H = L + \lambda^T f$, and the optimal control minimises H. This convention, used throughout Lenhart and Workman (2007) and Fleming and Rishel (1975), determines the signs of all subsequent control characterisations. The costate equations are $\lambda' = -\partial H/\partial y$, and the transversality conditions are $\lambda(T) = \partial\Psi/\partial y(T)$.

Hamiltonian.

The Hamiltonian H is constructed by adjoining the state equations (1)–(4) to the objective integrand via costate (adjoint) variables λ_S, λ_I, λ_F, λ_D:

$$H = \omega_D D + \omega_S S + (c_P/2)u_P^2 + (c_C/2)u_C^2 + \lambda_S(dS/d\tau) + \lambda_I(dI/d\tau) + \lambda_F\,(dF/d\tau) + \lambda_D(dD/d\tau) \tag{20}$$

Costate (adjoint) equations.

The backward costate equations are $d\lambda_i/d\tau = -\partial H/\partial y_i$ for each state variable $y_i \in \{S, I, F, D\}$. These are integrated backward in time from the transversality conditions at $\tau = T$. The terminal conditions are: $\lambda_{S(T)} = 0, \lambda_{I(T)} = 0, \lambda_{F(T)} = 0, \lambda_{D(T)} = \omega_T$ (the terminal damage weight from the objective functional, Equation 9).

Optimal control characterisation.

Derivation of u*c: The control u_C enters the state equations via $\partial(dS/d\tau)/\partial u_C = -S$ and $\partial(dI/d\tau)/\partial u_C = -I$ (Equations 1–2). Setting $\partial H/\partial u_C = 0$:

$$c_c\, u_c + \lambda_s(-S) + \lambda_i(-I) = 0$$

$$u_c = \frac{(\lambda_s S + \lambda_i I)}{c_c} c_c$$

With box constraints:

$$u^*_{c(\tau)} = min(uc, max, max(0, (\lambda_s S + \lambda_i I)/c_c)) \tag{21}$$

Sign interpretation: The costates λ_S and λ_I represent the marginal cost of an additional susceptible or parasitised larva, respectively. Under minimisation, these are typically positive ($\partial V/\partial S > 0$, since more larvae increase damage). Hence $(\lambda_S S + \lambda_I I)/c_C$ is positive, and u^*_C is active (positive) when the marginal benefit of removal exceeds the marginal cost.

Derivation of u^*_k: The control u_P enters the state equations via $\partial(dF/d\tau)/\partial u_P = +1$ (Equation 3: $dF/d\tau = \eta\delta I - \mu_F F + u_P$). Setting $\partial H/\partial u_P = 0$:

$$C_P\, u_P + \lambda_F \cdot (+1) = 0$$

$$u_P = -\lambda_F \,/\, c_P$$

With box constraints:

$$u^*_{P(\tau)} = min(u_P, max, max(0, -\lambda_F/c_P)) \quad (22)$$

Sign interpretation: Under minimisation, the parasitoid costate λ_F represents the marginal cost of an additional parasitoid. Since more parasitoids reduce beetle damage (a benefit), λ_F is typically negative ($\partial V/\partial F < 0$). Therefore $-\lambda_F/c_P$ is positive, making u^*_P active when augmentation is beneficial. **Note:** The previous version of this annex contained an error in Equation (22), writing λ_F/c_P instead of $-\lambda_F/c_P$. Under the minimisation convention $H = L + \lambda^T f$, the correct expression is $-\lambda_F/c_P$, which follows directly from $\partial H/\partial u_P = c_P u_P + \lambda_F = 0$.

Annual bird-habitat control.

The annual bird-enhancement control $u_B(t)$ is determined by the discrete optimality condition at the annual scale, embedded within the multi-year recurrence. Since u_B enters the model through Equation 7 and affects the within-season dynamics through the bird predation terms in Equations 1-2, its optimal value depends on the full annual cost-to-go and is updated at the annual (not within-season) timescale.

Numerical solution: Forward–backward sweep method (FBSM).

The resulting two-point boundary value problem is solved numerically using the forward–backward sweep method (FBSM), which iterates until convergence:

(i) Forward integration of the state system (Equations 1–4) with current control estimates from $\tau = 0$ to $\tau = T$.

(ii) Backward integration of the adjoint system ($d\lambda/d\tau = -\partial H/\partial y$) from $\tau = T$ to $\tau = 0$ using the terminal transversality conditions.

(iii) Update of the controls via Equations (21- (22), with convex combination damping: $u^{new} = \alpha\, u^{update} + (1 - \alpha)\, u^{old}$ (with $\alpha = 0.5$).

(iv) Convergence check: repeat until $\max|\Delta u| < \varepsilon$ (tolerance $\varepsilon = 10^{-4}$).

Convergence of the FBSM for systems of this structure is established by Hackbusch (1978) and applied in eco-epidemic contexts by Lenhart and Workman (2007, Chapter 11). The FBSM yields the optimal within-season profiles $u^*_P(\tau)$ and $u^*_C(\tau)$ for a single season. In the multi-year setting, the FBSM is applied sequentially within each year (receding-horizon / myopic strategy): the annual state x_t determines the initial conditions, the FBSM computes the optimal within-season controls, and the resulting terminal values determine x_{t+1} via the annual map. This sequential approach is distinct from solving the full N-year dynamic programming problem simultaneously, and the text reports results accordingly.

Step 9: Strategy comparison under optimal control

Method.

Steps 3–7 are repeated with the annual map G replaced by the controlled map G^{u^*}, where u^* is the optimal policy obtained in Step 8. The controlled fixed point x^{*ctrl}, controlled Jacobian $J_G(x^{*ctrl})$, and controlled tipping parameter p^{ctrl}_c are computed. The stabilisation gain from optimal control is quantified by the shift:

$$\Delta pc = pc^{ctrl} - pc^0$$

where p^0_c is the uncontrolled tipping boundary.

Control scenarios for hypothesis testing.

Scenario A: $u_P > 0$, $u_B = 0$, $u_C = 0$ (parasitoid augmentation only). Tests whether parasitoid attack alone is sufficient for system stability (H_{1a}).

Scenario B: $u_P > 0$, $u_B > 0$, $u_C = 0$ (combined parasitoid and bird-habitat enhancement). Tests whether bird predation provides essential complementary resistance (H_1^b).

Scenario C: $u_P > 0$, $u_B > 0$, $u_C > 0$ (full integrated control). Benchmark for the maximum achievable stabilisation.

Performance criteria.

Control performance is evaluated by four criteria applied to each scenario's long-run trajectory: (a) whether the equilibrium defoliation D* remains below D_crit = 0.5; (b) whether the equilibrium carrying capacity K* remains above K_min = 0.5 K_0; (c) the dominant eigenvalue magnitude ρ* at the controlled fixed point (resilience); and (d) the total multi-year cost J. Additionally, the Van Meerbeek resistance (R1) and resilience (R2) indices are computed for each scenario at the controlled fixed point.

Strategy decision rule. Strategies are compared using a lexicographic decision rule: (i) feasibility-only strategies satisfying D* < D_crit and K* > K_min and maintaining parasitoid coexistence (F* > 0) are considered; (ii) among feasible strategies, the one with minimum total cost J is selected as the minimal-effort strategy for each phenological scenario. H1 is supported if the most probable equilibrium across parameter space is the desirable coexistence equilibrium. If it is not, the analysis reports which phenological scenarios prevent convergence to the desirable state.

Purpose.

Assess whether the qualitative stability conclusions and hypothesis decisions are robust to uncertainty in parameter values and model structure.

Method 1: Latin Hypercube Sampling with Partial Rank Correlation Coefficients (LHS–PRCC).

Framework (Marino et al., 2008; Saltelli et al., 2008): Because several parameters are anchored by cross-system proxies rather than direct field measurements for the Beetle–M. mutabilis system (notably β, h, σ_A, σ_F), all stability and optimal control results are evaluated for robustness across the full plausible parameter ranges reported in the parameter tables (Sections 2.1.2–2.1.3 of the main text). Uncertain parameters are treated as intervals, and the analysis reports robust stability regions in parameter space—regions where the annual fixed point remains stable (all eigenvalues inside the unit circle) for all sampled parameter combinations.

Sensitivity of R_P, the tipping boundary p_c, and the optimal cost J to each parameter is quantified using partial rank correlation coefficients (PRCC) computed from Latin Hypercube Sampling (LHS). LHS stratifies each parameter's range into N equal-probability intervals and samples one value per interval with random permutation across parameters, ensuring uniform marginal coverage. PRCC then computes the correlation between each parameter and the output after removing the linear effects of all other parameters via rank regression, yielding a measure of monotonic sensitivity that is valid for nonlinear but monotonic input–output relationships.

A qualitative stability conclusion (i.e., which equilibrium is most probable, the regime shift probability, and the minimal-effort strategy ranking) is considered robust if it does not change across the full plausible parameter space.

Method 2: Structural robustness via alternative functional forms.

In addition to parametric sensitivity, structural robustness is assessed by comparing the baseline model against alternative functional form assumptions:

(i) Beverton–Holt versus Ricker recruitment in Equation (5): $S(0) = R_B A/(1 + A/K)$ versus $S(0) = A \exp(r(1 - A/K))$. This tests sensitivity to the form of density-dependent recruitment.

(ii) Holling Type II versus linear functional response for bird predation: $c_B B_t S/(1 + a_B(S + I))$ versus $c_B B_t S$. This tests sensitivity to the assumed saturation in bird foraging.

(iii) S → I compartmental parasitism versus direct parasitoid-induced mortality: the current model (parasitised larvae continue feeding until parasitoid emergence) versus a model where parasitism

directly removes larvae. This tests whether the explicit infected class is necessary for the qualitative conclusions.

If the qualitative stability conclusions (i.e., which hypothesis is supported and which bifurcation type governs tipping) remain unchanged under both parametric and structural perturbations, they are considered to reflect genuine properties of the tri-trophic interaction architecture rather than artefacts of specific modelling choices.

Model Adequacy and Data-Lean Design

The lack of long-term, site-specific time series for Beetle outbreak density, M. mutabilis parasitism prevalence, and passerine kill rates on this beetle does not constrain the class of results targeted by this study. The primary contribution of the model is to map the structural stability landscape by identifying parameter combinations and control allocations that yield a stable low-damage equilibrium, those that result in tipping, and the response of the parasitoid invasion threshold R_P to intraguild predation and control effort.

The focus is on structural stability properties (fixed-point existence, eigenvalue conditions, bifurcation type), which depend on the architecture of the annual map and the qualitative form of the functional responses, rather than on precise numerical parameter values. This approach aligns with standard practice in theoretical eco-epidemic modelling (Chattopadhyay and Arino, 1999; Hethcote et al., 2004), where the mathematical framework is constructed to be valid across the plausible parameter space.

To address the absence of precise parameter estimates, the analysis incorporates two complementary safeguards. First, all uncertain parameters are treated as intervals, as reported in the parameter tables of Sections 2.1.2–2.1.3 with empirically justified ranges, and stability and control results are presented as robust regions in parameter space rather than as point estimates. Conclusions are accepted only if they hold across the entire range of each uncertain parameter. Second, structural robustness is evaluated by substituting the baseline functional forms with ecologically plausible alternatives (Step 10). If the qualitative stability conclusions remain unchanged under both parametric and structural perturbations, they are considered to reflect genuine properties of the tri-trophic interaction architecture rather than artefacts of specific modelling choices.

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